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A novel three-phase target channel allocation scheme for multi-user Cognitive Radio Networks

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ABSTRACT

Generation and allocation of Target Channel Sequence (TCS) in relation to spectrum handoff introduces several challenges in a multi-user Cognitive Radio Network. The primary challenge is to ensure fair and unique TCS allocation to multiple Secondary Users (SUs) in order to prevent channel access conflict during concurrent handoff operations. Another challenge is to perform dynamic TCS generation to guard against channel obsolescence effects under time-varying user activities. Inclusion of real-time SUs further mandates the use of time-bound TCS generation in order to prevent call drops due to time-outs during handoff. To address these issues of fairness, uniqueness, dynamic and time-bounded TCS generation, this paper proposes a novel machine-learning enabled three-phase TCS generation and allocation scheme for multiple real-time SUs involved in Voice over IP (VoIP) communication. While proactive and periodic phases compute unique TCS in O (n log n) time, the reactive phase performs on-demand TCS generation. Generation of TCS is formulated as a fractional knapsack problem and greedy technique is applied therein to optimally select channels based on their existing conditions. A novel HMM-BiLSTM (Hidden Markov Model-Bidirectional Long Short Term Memory) framework is further devised to safeguard VoIP interests through periodic channel eligibility prediction. Additionally, an analytical framework is developed which records negligible probability of channel access conflict among multiple SUs. Simulation results confirm fair and dynamic channel allocation with reduced handoff delays and call-drops and intelligent learning based decision-making operations with high accuracy in comparison to existing approaches. Finally, test-bed implementation validates practical applicability of the proposed methodology.

1. Introduction

Cognitive Radio Network (CRN) [1–3] promises to alleviate the problem of spectrum congestion in next-generation networks through its unique underlay and overlay modes of communication. Specifically in the overlay mode, unlicensed Secondary Users (SUs) initiate spectrum handoff operations to switch to other available idle channels as soon as the licensed Primary Users (PUs) arrive in their designated channels. In order to preserve Quality of Service (QoS) for both PUs and SUs, the spectrum handoff procedure must incorporate two crucial functionalities. These include – (1) timely decision by SU to drop from the existing channel to prevent interference with incoming PU traffic; and (2) efficient target channel selection to minimize degradation during handoff. An SU may encounter several such instances of spectrum handoff. This necessitates the determination of a set of target channels, also termed as Target Channel Sequence (TCS) [4] which is also the focus of study in this paper.

Several works in literature [5–12] consider that SUs calculate their own target channels. This introduces additional constraints which include – (1) increased message overhead due to distributed mode of operation by SUs [5]; and (2) limited channel sensing capabilities of SUs [12]. On the contrary, delegating the task of TCS generation to a Spectrum Controller (SC) [13] has its advantages such as – (1) reduced handoff latency [8]; (2) reduced delay under a dedicated radio interface of SC [14] in contrast to a multi-transceiver based SU [10,15–17]; (3)overcoming the constraint of limited spectrum handoff agility in SU [18]; and (4) viability of practical implementation [19–21]. This paper, therefore, considers a centralized CRN where SC node is responsible for generation of TCS for multiple SUs and allocation operations.

In comparison to the TCS generation procedure for a single SU, addressing multiple SUs in CRN introduces additional complexities. The primary challenge is to ensure a "unique" sequence of target channels for each SU. Allotting the same set of target channels [4,22] may lead to channel access conflict when two SUs execute spectrum

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handoff concurrently. In this regard, unique TCS is generated for each SU in [23] using probabilistic TCS calculation method that minimizes the service and sojourn times. However, this work does not focus on the second challenge imposed by a multi-user CRN, which specifically refers to maintaining fairness during target channel allocation to multiple SUs. Notable works in [5,24] address this fairness issue. However, a distributed network is followed in [5] and is thus prone to the issues introduced by SU based channel selection as stated earlier. On the other hand, every SU initiates target channel selection in [24] through coordination with other SUs after being fed with channel sensing data by SC. In absence of TCS, this may result in increased overhead due to cooperation and associated delays, rendering it unsuitable for real-time applications.

Dynamic network conditions impose another challenge during TCS generation and allocation. This is because channel obsolescence effects render proactive mode of TCS generation [4–6,15,16,21–23,25–31] unsuitable under diverse channel conditions. Although reactive approach [32–35] solves this issue through on-demand channel sensing and selection, it leads to increased handoff delays. This delay imposes additional constraint when real-time SUs are involved in a multi-user CRN. For example, higher handoff delays during Voice over IP (VoIP) sessions may lead to call time-outs and hence, call drops when RTCP (Real-Time Transport Control Protocol) packets are not received within a threshold time [36].

This work considers such real-time SUs that are engaged in VoIP communication. Increased popularity of low-cost VoIP technology (where voice is transmitted as IP packets) [37] makes it a perfect candidate for CRN [38–40] and is thus selected for this work. However, strict QoS guarantees for VoIP imply that channel consensus during spectrum handoff must be attained within the RTCP call time-out period. Since channel pre-emption is not possible between two VoIP SUs having equal priorities, the TCS generation method must be adaptive to network conditions and periodically updated to prevent collision among multiple SUs. This additional constraint of QoS makes this work even more challenging.

Currently application of ML (Machine Learning) techniques for this domain is being encouraged due to their learning capability under dynamic conditions. However, such works are only limited and emerging [41–43]. A recent work in [43] proposes the use of Q-learning technique for spectrum handoff supported by test-bed execution. This provides a strong motivation for developing practically feasible ML solution in this paper. However, there is a tradeoff between accuracy and timeliness while building such models. Specifically with respect to this paper while CRN requires accurate prediction of network states, VoIP demands timely decision-making solutions. Both these factors are not yet explored jointly and this provides impetus for further exploration in this domain.

All the aforementioned challenges are successfully addressed in this paper. Specifically, the objective is to devise and execute a fair TCS generation and allocation scheme catering to multiple real-time SUs under dynamic network conditions. To this end, the significant contributions of this paper are enlisted as follows.

i. Formulation of a novel three-phase scheme that generates and periodically updates time-bound unique TCS sets and further allocates the same to multiple VoIP SUs under time-varying PU traffic.
ii. Modeling of a joint HMM-BiLSTM (Hidden Markov Model-Bidirectional Long Short Term Memory) framework in accordance with the proposed three-phase scheme for accurate as well as real-time prediction of channel conditions based on application requirements of VoIP SUs.
iii. Test-bed design and practical implementation of TCS generation and allocation by SC with incorporation of the ML framework and subsequent spectrum handoff operations by multiple VoIP SUs.

The underlying analytical framework as devised in this paper confirms both fair channel allocation as well as avoidance of channel access conflict during spectrum handoff in a multi-user CRN. Additionally, the proposed method succeeds in sustaining VoIP calls with adequate QoS even under increasing PU traffic. In comparison to similar works in literature, both call drop probability and handoff delay are drastically reduced in this work. Finally, complete test-bed implementation of the proposed solutions is demonstrated for the first time in this paper to the best of our knowledge.

The proposed methodology further provides thrust to new research areas of spectrum handoff with added features, specifically targeting – (1) *hybrid spectrum handoff* strategy by SUs using a combination of proactive and reactive approaches; (2) *channel modeling through learning* from past handoff experiences; (3) *traffic classification and formulation of MAC protocols* based on connection disruptions and handoff instances; (4) *test-bed based prototype design of SC* with focus on handoff based channel reservation and call admission control, etc.

The rest of the paper is organized as follows. Section 2 establishes the novelty of our work through an extensive literature survey. Section 3 introduces the system model. The proposed methodology is described in detail firstly with respect to the three-phase scheme in Section 4 followed by the proposed HMM-BiLSTM framework in Section 5. Section 6 executes performance evaluation in simulation which is validated through test-bed implementation in Section 7. Finally, the paper is concluded in Section 8.

2. Related works

TCS calculation and allocation methodology in this paper focuses on four significant aspects. These include – (1) TCS for multiple SUs; (2) TCS for real-time SUs; (3) TCS under dynamic network conditions; and (4) ML-empowered TCS. This section provides an outlook of ongoing research in these domains and subsequently highlights the novelty of the work in this paper.

2.1. Literature survey – TCS for multiple SUs

The issue of multiple SUs is mostly addressed through user classification. Several grade-of-service based approaches [44–49] prioritize users as PUs, real-time SUs and non real-time SUs. However, there are certain unresolved issues in these works. While [44] assumes a fully connected CRN, [46] performs channel preemption to avoid call drops. On the other hand, spectrum holes reserved for real-time SUs in [45] are not explored in detail. In general, the scenario of multiple real-time SUs accessing the same set of available channels during handoff is not yet discussed. Moreover [48,49] focus on single target channel allotment instead of TCS generation. Finally, spectrum leasing is considered in [47] which is fundamentally different from the concept of opportunistic access in this paper. It must be noted that multi-SU based works which focus on either distributed CRN [5,6,9,11,12,26,50] or allotment of single target channel [16,35,51–54] (in contrast to TCS) are omitted from further discussion in this paper.

Another notable work in [23] uses probabilistic approach for TCS generation such that SUs exploit different sequences. However, this work refrains from any discussion on fairness or allocation aspects. Additionally, the concept of using two transceivers for every SU [15,23] may not be a cost-effective or energy-efficient solution. Similarly, having backup channels for handoff [55] is not always feasible under dynamic channel conditions. On the contrary, this paper focuses on TCS generation and *conflict-free, fair* allocation to *multiple SUs of same priority* without *any added transceiver or backup channel.*

2.2. Literature survey – TCS for real-time SUs

Real-time SUs demand generation of time-bound TCS. As acknowledged in [56], there is a trade-off between optimal channel selection and computational complexity during TCS calculation. To cite an instance, dynamic programming based optimal TCS generation in [22] incurs high computational complexity of $O(LM^2)$ where M and L denote the number of candidate channels and target channels in TCS respectively. The complexity is reduced to O(M) time using greedy approach [22,27,57] that decreases the number of TCS comparisons. However, this does not guarantee optimal sequence for SUs. Likewise, [23] formulates TCS design as a convex optimization problem so as to reduce complexity overhead, resulting in suboptimal solutions. Deviating from TCS calculations, [58] proposes an optimal channel sensing sequence that reduces handoff delay at the cost of increased time complexity $O(M^3)$. It is therefore understood that generation of time-bound as well as optimal TCS for real-time SUs is still an open research problem.

From the user perspective, significant works focus on QoS aware audio/video applications [8,12,24,59] besides priority based user classification [45,46]. Specifically, [8] achieves lower handoff delay under negotiated spectrum access compared to opportunistic access. However, negotiated access is not possible when SUs are anonymous and independent in their operations. On the other hand, handoff delay of 20 s in CR test-bed [59] and 350 ms under relay protocol [60] are not acceptable for real-time SUs. Additionally, [12] assumes that SUs have knowledge of PU arrival rate in each channel. This may not be feasible in a generic CRN. On the other hand, SU's incapability to sense more number of channels [12] can be addressed by deploying SC terminal as stated earlier. Although SC nodes are used in [12], two issues demand our attention. Firstly, target channel selection is initiated by SU using cooperative approach. This implies that during channel access conflict, this cooperative method must be repeated, incurring higher delays. Secondly, in absence of TCS generation or allocation process, it is not clear how multiple channel/user status is maintained in this work.

In terms of classification, the work in [45] uses best, worst and first match techniques for channel selection. In absence of TCS, such arbitrary selection may increase channel access conflicts whereas the use of backup channels in [49] may lead to failure due to channel obsolescence effects.

Another category of works emphasize on minimizing handoff and sensing delays. For example, handoff reduction is done using backup channels [55], fuzzy logic [17,49], LMS algorithm with adaptive weights [61] and grade-of-service based approaches [44–47]. Steady state analysis of handoff delay and channel availability is further conducted in [62]. On the other hand, different channel sensing sequences [7,12,53,58,63] aim at reducing sensing delays. However, these works focus more on handoff issues without elaborating on TCS generation and allocation schemes. Finally, it is inferred in [11,58] that the "always switch" policy incurs higher delays than the "stay-on" policy due to additional sensing and switching delays during every channel switch. On the contrary, it is more feasible for real-time SUs to perform channel switch to avoid time-outs and call drops. Accordingly, this paper generates *optimal time-bound* TCS with *reduced handoff delay guarantees* and "*always switch" policies* for real-time SUs.

2.3. Literature survey – TCS under dynamic network conditions

Two performance metrics can effectively evaluate the varying network conditions. These include (1) link maintenance probability [13,47,64–66]; and (2) connection disruption rates [3,28,31]. Link maintenance under dynamic network conditions is improved using backup channels [64,66] and adaptive power control techniques in distributed [47,64] or centralized networks [65]. While [47] relies on certain non-generic considerations to apply fuzzy logic, [65] does not explore TCS calculation or allocation issues. Moreover, adaptively controlling SU transceiver properties require consensus with other SUs in a multi-user CRN. This may incur additional delays rendering this scheme inappropriate for VoIP traffic. Backup channels, on the other hand, are prone to channel obsolescence effects under varying conditions.

Another category of studies [3,31] performs probabilistic estimation of remaining time periods and channel conditions to reduce connection disruption during handoff. SUs are equipped with channel usage information which aid in estimation and decision-making. Therefore, success of these methods relies on their estimation procedure which may underperform under degraded channel conditions. Besides, dynamic channel selection method in [28] advocates the use of "stay on the current channel" policy which is not favorable for real-time SUs as mentioned earlier. In contrast to these works, our paper focuses on a "*three-phase*" approach for *adaptive generation of TCS without use of adaptive power control, backup channels, or prior channel usage information*.

2.4. Literature survey – TCS powered by ML techniques

ML techniques for CRN are only recently being reported. Mostly these works focus on spectrum sensing [67–70] with primary attention on PU classification [71,72] using supervised and deep learning methods. On the contrary, only few studies [73,74] have explored spectrum management and mobility issues in CRN [41–43]. Specifically, [41] has devised supervised learning method to assist in handoff using Particle Swarm Optimization technique. Swarm intelligence is used in [42] by exploiting the social intelligence parameter in spectrum sharing. However, these works rely on cooperative spectrum sensing and management. This paper, on the contrary, delegates spectrum management and decision process to the SC in order to relieve real-time SUs from the complexity of ML.

Finally, both the works in [43,75] establish the credibility of implementing practically feasible ML solutions for CRN and serve as a strong motivation for this work.

However [75] does not focus on SC–SU interaction or TCS generation. On the other hand, Q-learning based reinforcement learning method for enabling cognitive handoff is evaluated experimentally in [43]. Every SU in [43] learns from an expert node through Q-tables as the network evolves. It requires at least 30 s and 10 s under "RL" and "TL" methods in [43] for SUs to achieve the "optimality" condition. QoS aware SUs in our paper may not allow this relaxation in learning time as otherwise call quality shall drop leading to call drops. It is to be realized that while model building/training time may incur few seconds, channel prediction for VoIP SUs must occur at least within the order of μs and the same is addressed in our work. Specifically, this paper focuses on a *practically feasible HMM-BiLSTM based channel prediction method* to assist in *quick prediction of target channel condition for every SU*.

In a nutshell, the novelty of this work is ascertained by the fact that the proposed scheme caters to *ML-enabled adaptive TCS generation and allocation for real-time SUs in a multi-user CRN under diverse network conditions,* and further validates the same through *complete test-bed design and implementation*.

3. System model

We consider an infrastructure-based overlay CRN consisting of N_s SUs and N_p PUs. Each PU is assigned a non-overlapping channel, for a total of N_p channels. All SUs are under the control of a Spectrum Controller (SC) which is also connected to the VoIP proxy server. SC is equipped with two transceivers and is involved in TCS calculation and allocation for its SUs. Taking into account frequency agility threshold of SUs [18], SC senses W channels ($W \leq N_p$). SC allocates a default idle channel (if available) to SU. SU performs spectrum sensing using energy-detection method [3] and resumes transmission only if the channel is sensed idle. This listen-before-talk model is adopted from IEEE 802.22 standard [76].

Table 1
TCS generation under Proactive, Reactive, and Periodic Phases.

	Proactive	Reactive	Periodic
Activated:	After successful handoff by each SU	When all channels in pre-determined TCS are sensed busy by SU during handoff	At regular intervals during SU transmission
Purpose:	Update TCS when channel availability status changes after each handoff operation by SU	Update TCS when channels in the pre-determined TCS are currently unavailable for SU access	Update TCS to guard against channel obsolescence effects in dynamic network conditions
Methodology:	Greedy solution to fractional knapsack based problem	On-demand pseudorandom sequence	–Same as Proactive–
Properties:	Optimal and fair channel allocation, low time-complexity, collision avoidance	Collision avoidance, fair channel allocation	–Same as Proactive–
Priority:	Lower than Reactive, higher than Periodic	Highest	Lowest
Frequency of occurrence:	Depends on the no. of handoff operations	Minimum; occurs only under the Worst-case scenario	At periodic intervals depending on no. of active SUs

Table 2
List of abbreviations.

PU	Primary User	l_pro	Proactive TCS message
SU	Secondary User	l_react	Reactive TCS message
SC	Spectrum Controller	C_V	Channel Allocation Vector
CRN	Cognitive Radio Network	U_V	User Allocation Vector
CCC	Common Control Channel	T_V	TCS Allocation Vector
TCS	Target Channel Sequence	RT	Real-Time users (VoIP SUs)
PROLOC	Proposed Methodology	CSP	Channel State Prediction block

VoIP SUs are marked as real-time (RT) users to prioritize their transmissions over non real-time users. SC serves RT SUs in a First Come First Serve (FCFS) Policy [4]. Unless explicitly stated, any SU henceforth mentioned in this paper refers to a VoIP SU. Also similar to the work in [22], only one VoIP call is allowed in the allotted channel. However, SU must relinquish access to this channel and perform spectrum handoff as soon as the designated PU arrives for transmission.

During spectrum handoff, SC has to ensure that at a given time, any two interrupted SUs must select different target channels for resuming transmission. SC deploys one transceiver for channel sensing. It calculates idle-to-busy and busy-to-idle transition rates for each channel and estimates its idle and busy periods. SC then creates a pool of sensed channels and calculates unique TCS for every SU as well as generates the Common Control Channel (CCC). Using the second transceiver, it broadcasts this TCS to SU using CCC.

Each TCS includes the sequence of target channels and their estimated busy periods. During handoff, the SU selects a target channel from its TCS. It waits for the busy period associated with the channel. Thereafter, the SU senses the channel and resumes transmission only if the channel is sensed idle. In this paper, the proposed TCS calculation and allocation methodology involves three phases. A comparative study of these three phases is provided in Table 1 and is discussed in the following sections. A list of abbreviations used in this paper is further summarized in Table 2.

3.1. Proactive phase

This phase is activated as soon as SU switches to a target channel following spectrum handoff. SU informs SC of the newly selected channel via *l_pro* message. SC generates the proactive TCS and sends it as an acknowledgment to the concerned SU. During TCS calculation, SC applies greedy technique to the created channel pool by formulating the channel calculation problem as a fractional knapsack problem.

3.2. Reactive phase

In the worst-case scenario, an SU after dropping from its current channel may sense all the listed target channels busy in its TCS. In order to resume VoIP call at the earliest, SU seeks for available target channel by sending *l_react* message to the SC. SC generates on-demand reactive TCS and notifies the SU of the same in reply. After successful handoff to a target channel, SU again sends *l_pro* message to the SC and repeats the procedure of the proactive phase as mentioned earlier.

3.3. Periodic phase

This phase is in effect during normal SU transmission. SC periodically updates the TCS and informs SU via its operating channel. The methodology for TCS calculation remains the same as discussed in proactive phase. It is worth mentioning that the proactive phase updates the channel list on account of changes in SU activity. On the contrary, periodic TCS updates the TCS to guard against channel obsolescence effects.

In order to prevent the worst-case reactive scenario from occurring frequently, SC is further equipped with an intelligent Channel State Prediction (CSP) block. This block implements a novel joint HMM-BiLSTM based ML procedure that scans every operating channel under SU transmission and predicts any adverse channel condition. SC sends *ch_react* message to any SU for which the operating channel is deemed unfit, thereby initiating spectrum handoff by SUs. The idea is to compel SUs into performing handoff so that they can receive updated periodic TCS and avoid reactive phase. This not only improves overall QoS but also reduces the number of call drops that may otherwise occur during the reactive phase.

Illustrative Example: We consider a CRN with 3 SUs and 11 idle channels as depicted in Fig. 1. SUs are either transmitting or in handoff mode at different timestamps. Based on PU arrival, SC must allocate suitable TCS to SUs under the proposed scheme. To this end, SC generates periodic, proactive and reactive TCS empowered by the ML method and SUs operate accordingly. Their actions are explained as follows.

i. At time t_o: SU_1, SU_2 and SU_3 are transmitting in channels Ch_1, Ch_2 and Ch_3 respectively. SC updates each SU with periodic TCS. Let the TCS sets for SU_1, SU_2 and SU_3be $\{Ch_4, Ch_5\}$, $\{Ch_5, Ch_7\}$, and $\{Ch_7, Ch_9\}$ respectively.
ii. At time t_1: A licensed PU arrives in Ch_1. SU_1 performs handoff and switches to the first target channel Ch_4. SU_1 informs SC via the new channel with *l_pro* message. SC selects the channel pool $\{Ch_8, Ch_9, Ch_6, Ch_{10}\}$ out of the available channels for generation of the updated TCS.
iii. At time t_2: SC generates and sends proactive TCS $\{Ch_8, Ch_9\}$ to SU_1. It updates and sends periodic TCS sets $\{Ch_9, Ch_6\}$ and $\{Ch_6, Ch_{10}\}$ to SU_2 and SU_3 respectively.

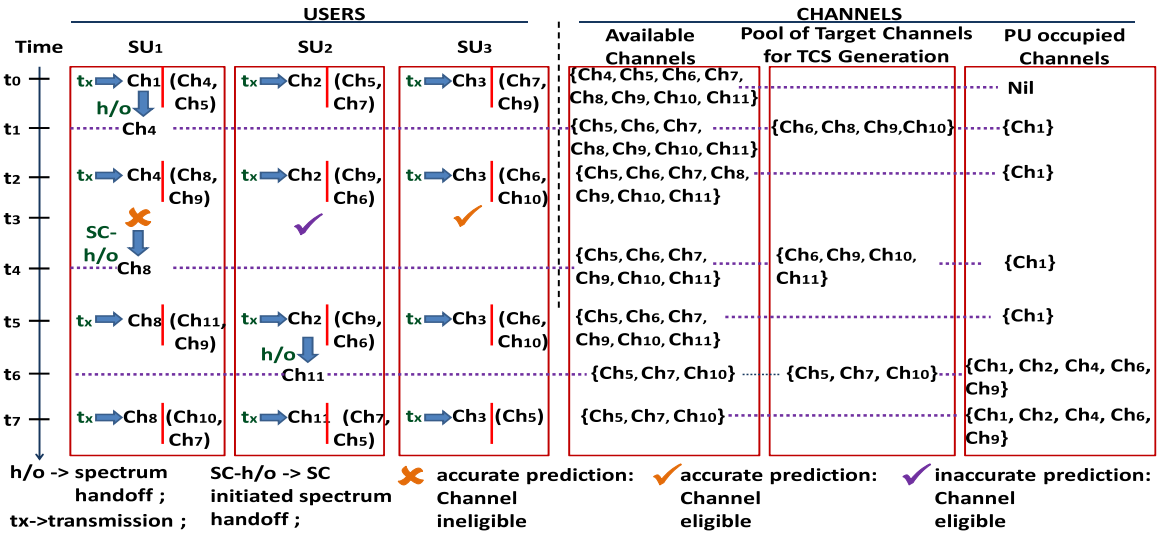


Fig. 1. Illustrative example of TCS generation and allocation to multiple SUs under proactive, reactive and periodic phases powered by ML-based channel prediction.

iv. At time t_3: SC predicts channel conditions for all active channels Ch_4, Ch_2, and Ch_3. It accurately predicts Ch_3 as eligible and Ch_4 as ineligible for transmission. Ch_2 is also ineligible. However SC misses prediction for Ch_2. It thus sends *ch_react* message only to SU_1 (omitting SU_2).
v. At time t_4: SU_1 performs SC initiated proactive handoff to its target channel Ch_8 and follows the procedure as in t_1.
vi. At time t_5: SC updated TCS for SU_1, SU_2 and SU_3 are $\{Ch_{11}, Ch_9\}$, $\{Ch_9, Ch_6\}$ and $\{Ch_6, Ch_{10}\}$ respectively.
vii. At time t_6: 4 more PUs arrive in channels Ch_2, Ch_4, Ch_6 and Ch_9. SU_2 senses all channels busy in its periodic TCS $\{Ch_9, Ch_6\}$. SU_2 finds all channels busy in its TCS. It seeks target channel by sending *l_react* message to SC. SC performs a pseudo-random sequence of available channels $\{Ch_5, Ch_7, Ch_{10}, Ch_{11}\}$ and sends this reactive TCS to SU_2. SU_2 randomly selects Ch_{11} and resumes transmission as Ch_{11} is idle. SC selects available channel pool $\{Ch_5, Ch_7, Ch_{10}\}$ for further update of TCS.
viii. At time t_7: SC finally updates periodic TCS $\{Ch_{10}, Ch_7\}$, $\{Ch_7, Ch_5\}$ and $\{Ch_5\}$ for SU_1, SU_2 and SU_3 respectively.

SU_1, SU_2 and SU_3 represent user actions in proactive, reactive and periodic phases respectively. Notably ML-enabled CSP block of SC prevents SU_1 from reactive TCS by initiating handoff earlier than PU arrival. Another observation from Fig. 1 is that even though SU_3 has not encountered a single handoff, its TCS is still updated because of changing network conditions. Thus TCS is immune to channel obsolescence effects. These aspects are discussed in detail in the following sections.

4. Proposed adaptive TCS generation and allocation scheme

In this section, the proposed three-phase scheme is studied in detail with its merits and challenges. Inclusion of CSP further enhances SC functionality and is discussed in Section 5.

There are two prerequisites to this three-phase scheme, which include – (1) generation of channel pool for TCS calculation; and (2) maintaining different allocation vectors. These are discussed in the following.

4.1. Prerequisites for the proposed TCS generation and allocation scheme

In order to generate the pool of target channels, SC deploys one of its two transceivers to sense each channel w for a total of W channels. Subsequently, channel conditions are derived based on the work in [77]. Under exponential on–off traffic at the PU end, the effective and non-effective communication durations for a particular channel as denoted by T_{EFF} and T_{N-EFF} respectively are given by the following [77].

$$T_{EFF} = \frac{1}{\beta}\left(1 - e^{-\beta T_{SI}}\right) \tag{1}$$

$$T_{N-EFF} = \frac{1}{\alpha}\left(1 - e^{-\alpha T_{SI}}\right) \tag{2}$$

where, α, β are the transition rates from busy to idle and idle to busy periods respectively, and T_{SI} is the inter-sensing duration.

SC selects eligible channels for TCS calculation based on frame duration value (denoted by T_v) of the admitted SUs. To this end, SC compares T_{EFF} of each channel with T_v and allows only those channels for TCS calculation which allow transmission of a predefined number of VoIP frames without disruption. This number depends on the VoIP codec and is well documented in [78]. Being connected to the VoIP proxy server as per the system model, SC is informed of the codecs used and it accordingly prepares the list of eligible channels.

It must be noted that determination of T_v for VoIP SUs is a challenging task for SC because of the complexities involved in VoIP traffic modeling. Moreover, this estimation may vary from actual transmission period incurred by SUs (denoted by T_{su}). Consequently, the proposed HMM-BiLSTM model in Section 5 helps to recover from any discrepancy between T_v and T_{su} by incorporating call quality along with channel state. T_v and T_{su} are derived in Appendices A and B respectively.

The non-effective communication period during spectrum handoff further includes channel switching time (t_{sw}), consensus time (t_c) and target channel sensing time (t_s), and is given by,

$$T_{N-EFF|HO} = \frac{1}{\alpha}\left(1 - e^{-\alpha T_{SI}}\right) + t_{sw} + t_c + t_s \tag{3}$$

where t_{sw}, t_c and t_s are explained in the following.

4.1.1. Determination of t_c, t_{sw}, t_s

Consensus time t_c is defined as the time taken by SC and SUs to have a consistent view of channel state before initiating transmission in the corresponding channel. This is included in $T_{N-EFF|HO}$ because an idle channel can only be effectively used after all stakeholders (SC and SUs) agree transmission on it.

t_c is therefore given by,

$$t_c = t_{RTS} + \left(\Delta t_{c1} + \Delta t_{c2}\right) + t_{CTS} \tag{4}$$

where Δt_{c1} = C_V search time (explained below); Δt_{c2} = propagation time for operating channel message on CCC; t_{CTS} = RTS-CTS message exchange; t_{RTS} is SU waiting time and is defined in Section 4.

Channel switching time t_{sw} is given by the time taken by SU to switch from the current operating channel to the new target channel. While this time is primarily hardware dependent, SC can determine this value indirectly. As per the proposed policy, SC is connected with VoIP proxy server. Hence during channel switch as RTP connection is paused, SC records the timestamp (denoted by $timestamp_{last_RTP}$). Similarly SC records another timestamp (denoted by $timestamp_{l_pro}$) of receiving the *l_pro* message which indicates successful switch to the new channel. t_{sw} is, therefore, simply the difference of these two timestamp values from the SC perspective and is given by,

$$t_{sw} = timestamp_{l_pro} - timestamp_{last_RTP} \tag{5}$$

Finally, t_s is the time incurred for sensing by SU as per its alternate sense and transmit strategy. Again SC can indirectly determine this value by utilizing its vectors. As every SU in VoIP session must acknowledge receipt of handoff using *l_pro*, SC can record the timestamp (denoted by $timestamp_{last_l_pro}$) for which the last remaining SU from the session transmits *l_pro* message. This can be verified using *C_V* and *U_V* vectors (which are explained below). Finally SU after completing its sensing operation resumes VoIP calls by initiating the RTCP control message that includes sending and receiver reports. SC records the corresponding timestamp (given by $timestamp_{first_RTCP}$) and uses the following expression to determine t_s.

$$t_s = timestamp_{first_RTCP} - timestamp_{last_l_pro} \tag{6}$$

SC estimates T_{EFF} and $T_{N-EFF|HO}$ values to rank each channel and creates a pool of target channels based on their ranks. We denote this channel pool using the notation W_T. Any idle channel is discarded from W_T whenever $T_{N-EFF|HO}$ is greater than or is equal to T_{EFF}. In order to resolve channel access conflicts during multiple handoff operations in a multi-user CRN, SC designates each VoIP session with a unique session ID and also maintains the following three allocation vectors.

(i) *Channel Allocation Vector C_V*: *C_V* maintains channel availability status of all channels in the target channel pool and is defined in the following.

$C_V = \{(w_i, v_i) : w_i \epsilon W_T\}$
where, $v_i = 0$ if w_i is not allotted to any SU;
$v_i = 1$ if w_i is chosen as the first channel in TCS;
$v_i = 2$ if w_i is selected as CCC;
$v_i = k$ if w_i is allotted to SUs under session ID = k

(ii) *User Allocation Vector U_V:* Let U denote all the SUs currently registered under an SC node. *U_V* therefore maintains session IDs of all these SUs and is defined as,

$U_V = \{(u_i, x_i) : u_i \epsilon U\}$
where, $x_i = k$ if u_i belongs to kth session ID.

(iii) *TCS Allocation Vector T_V:* *T_V* updates the TCS list and session ID for the subscribed SUs. It is defined as,

$T_V = \{\{(t_{i,1}; t_{i,2}), u_i\} : u_i \epsilon U\}$
where, $t_{i,1} = k$ if u_i belongs to the kth session ID;
$t_{i,2} =$ TCS set currently assigned to u_i

Equipped with the target channel pool W_T and allocation vectors, SC generates and allocates the TCS to multiple SUs in three phases. These phases are discussed in the following.

4.2. 1st phase — proactive TCS generation and allocation phase

This is the first phase of the proposed three-phase scheme. The calculation of proactive TCS must fulfill two conditions.

(i) The average handoff delay d incurred during every instance of spectrum handoff must not exceed a threshold limit D ($D \sim$ 1 s) or else the call is dropped due to time-out. Notably, CISCO specifies 1 s as minimum call time-out value for VoIP calls [36]. It is noteworthy that this paper imposes a more stringent threshold value as compared to other works in literature. To cite an instance, a threshold value of 18 s is adopted in [8] which is not applicable for VoIP calls.

(ii) The previous constraint imposed on d implies that the TCS calculation method must be time-bound. In other words, the method must not incur high computational complexity.

Under these two conditions, this paper formulates the TCS calculation problem as a fractional knapsack problem [79]. This enables the application of time-bound greedy technique for target channel selection as described in the following.

4.2.1. Greedy technique for proactive TCS generation

Knapsack problem [79] is defined as follows – "Given a sack of capacity C, how many items 'i' each of size S_i can be included to maximize the value of sack, where each item 'i' has a value v_i?". Fractional knapsack further allows a fraction of items in sack and is used to map the proactive TCS calculation problem with minor modifications as follows.

"Given the time limit of D, how many channels 'w_i' each having estimated non-effective communication slot $T^{(w_i)}_{N-EFF|HO}$ can be included in TCS so that cumulative value is maximized where each channel w_i is valued by effective communication time $T^{(w_i)}_{EFF}$?". Let W_T include n channels where $W_T = \{w_1, w_2, \ldots, w_n\}$. The objective function is formulated as follows.

$$\text{Objective Function: } \max\ f(j) = \sum_{j=w_1}^{w_n} \left(T^{(j)}_{EFF} \times x(j)\right)$$

$$\text{Subject to } \sum_{j=w_1}^{w_n} \left(T^{(j)}_{N-EFF|HO} \times x(j)\right) \le D \tag{7}$$

In order to solve this function, SC calculates the ratio of effective to non-effective communication duration for each channel in W_T. All the channels are then sorted based on their decreasing ratio values. After the sorting operation, each channel from the sorted list is included in the TCS set as long as the total non-effective communication duration is less than or equal to D. This fulfills the first condition of threshold delay limit as stated earlier in this section.

Mathematically, $x(j)$ denotes inclusion of the jth channel in the TCS and is given by the following.

$$\left.\begin{array}{ll} x(j) = 1 & \forall\ w_1 \le j \le w_{k-1} \\ x(j) = 0 & \forall\ w_{k+1} \le j \le w_n \\ x(j) = \dfrac{\sum_{i=w_1}^{w_k} T^{(i)}_{N-EFF|HO} - D}{T^{(j)}_{N-EFF|HO}} & \forall\ j = w_k \end{array}\right| \tag{8}$$

where, w_k is the final channel included in the TCS for a particular SU and is given by (9) and (10) in the following.

$$\frac{T^{(w'_1)}_{EFF}}{T^{(w'_1)}_{N-EFF|HO}} \ge \frac{T^{(w'_2)}_{EFF}}{T^{(w'_2)}_{N-EFF|HO}} \ge \cdots \ge \frac{T^{(w'_n)}_{EFF}}{T^{(w'_n)}_{N-EFF|HO}} \tag{9}$$

$$w_k = \min\left[w_i \in \{w'_1, w'_2, \ldots, w'_n\} : \sum_{l=w'_1}^{w_i} T^{(l)}_{N-EFF|HO} > D\right] \tag{10}$$

Eq. (9) refers to the situation when channels are sorted in terms of decreasing ratio values. Channels from this sorted list $\{w'_1, w'_2, \ldots, w'_n\}$ are included in the generated TCS. Once delay threshold is reached, the corresponding channel w_k is included as the last one in TCS as denoted by (10). In this case, only fraction of the valid time is used to calculate $x(j)$.

In this way, SC generates the proactive TCS for every SU. Incorporation of D ensures that even if SU selects the final channel from TCS, its waiting time is still below the call time-out limit. Once D is reached,

SC terminates the algorithm and sends this TCS to the SU. During subsequent TCS generation for the next batch of users, SC discards the first target channel included in the TCS of every other SU. This is done to reduce collision among these SUs during spectrum handoff.

Once the TCS is generated, the next step is to allocate this TCS to the SUs which is described in the following.

4.2.2. Allocation of proactive TCS

On receipt of *l_pro* message, SC broadcasts proactive TCS not only to the requesting SU but also to other SUs belonging to that VoIP session. It also updates all three allocation vectors.

During handoff, attaining consensus among SUs in a VoIP call is critical. This is because any delay may lead to call time-out. To this end, two possible scenarios are studied as follows.

(i) Scenario 1: *SU handoff in the same PU zone*

In this scenario, SUs involved in VoIP session are operating in the same transmission zone of PUs. As channel conditions are identical for both users, they drop from the channel almost concurrently. They scan each channel in TCS. When any one SU selects a target channel, it sends RTS message twice, each after a span of t_{RTS} period. The other SU with all probability also selects the same target channel (as channel conditions are identical). It receives at least one RTS message. It responds with CTS message and the call is resumed.

(ii) Scenario 2: *SU handoff in different PU zones*

Here the two SUs namely SU_1 and SU_2 reside in different zones with respect to PU transmission. This implies that while channel Ch_1 is sensed busy by SU_1 due to PU activity in zone 1, SU_2 detects Ch_1 as idle in a different zone (zone 2). Therefore, SU_1 performs handoff whereas SU_2 stays on the current channel. Their actions are discussed as follows.

- SU_1: It drops from Ch_1, selects a target channel Ch_2 from its TCS, performs handoff and informs SC of its new operating channel using *l_pro* message. It continues to transmit RTS messages to establish communication with SU_2.
- SU_2: SU_2 stays in Ch_1 and waits for t_{RTS} time. As there is no RTP/RTCP message from SU_1within this time, SU_2 moves to CCC for further information from SC.
- SC: It provides updated proactive TCS to both SUs. At the same time, it notifies the new operating channel Ch_2 to the listening SU_2on CCC using the session ID in *U_V* vector.

SU_2 now switches to Ch_2. It performs RTS-CTS message exchange with SU_1 and successfully resumes the call. Notably, SU_2 waits in the CCC for a maximum 2**D* time after which the call is dropped if there is no incoming message.

4.2.3. Analysis of proactive TCS

This subsection performs analysis of proactive TCS based on the properties stated in Table 1. Specifically, analysis is done in terms of optimal channel selection, time-complexity and collision avoidance during channel access by multiple SUs.

(i) Optimal selection of target channels

Let $TCS_{pro} = \{g_1, g_2, \ldots, g_k\}$ be the generated proactive TCS. For each channel g_i in TCS_{pro}, let value and weight refer to $T_{EFF}^{(g_i)}$ and $T_{N-EFF|HO}^{(g_i)}$ respectively. Further, we consider that an optimal solution O exists for the formulated fractional knapsack problem P where $O = \{o_1, o_2, \ldots, o_k\}$. Proof for optimal channel selection is to derive equivalence between O and TCS_{pro} by validating the following two properties.

- *Proof of greedy choice property*–There exists an optimal solution to P that selects the first greedy item g_1 from TCS_{pro}.

We assume O does not have g_1. Weight of g_1 is hence removed from O. g_1 is now added to form a new solution O'. O' fulfills the constraint imposed by D and g_1 has the best value/weight ratio. Hence O' is as good as O in giving the maximum cumulative value (or else there is a contradiction). It is thus inferred that $g_1 \epsilon O'$ and O' is optimal solution for P.

- *Proof of optimal substructure property* – Optimal solution to problem P comprises of optimal solutions generated from its subproblems.

As g_1 is removed from P in the previous step, P'' is the updated problem with the constraint $D'' = D - T_{N-EFF|HO}^{(g_1)}$. Let $O'' = O' - \{g_1\}$. The objective is to prove that O is an optimal solution to P. Let the optimal solution to P'' be Q and not O'' where Q has more value than O''. Furthermore, let $R = Q \bigcup \{g_1\}$. Obviously, R is a feasible solution to P. On the other hand, O' (optimal solution for P) and $O'' + \{g_1\}$ have the same value which is less than R's value. This is because value of O'' is less than that of Q. Thus, R becomes more valuable than optimal solution O' (which is a contradiction). Hence, it is inferred that O'' is an optimal solution to P''.

Using these two properties, it is proved inductively that after SU greedily selects a channel from TCS_{pro} as an optimal solution, the new problem has same form as the original one. This guarantees optimal target channel selection under proactive TCS.

(ii) Time-complexity analysis

The time complexity incurred during greedy selection of channels after sorting them is O (n log n) where n is the number of channels in W_T. This can be derived as follows.

- *Proof of time complexity:* There are three significant steps in the proposed algorithm namely – (1) Calculation of weight/value ratio for each channel that incurs a single loop spanning n number of channels; (2) Sorting of channels in terms of their ratios using quick sort procedure; and (3) TCS generation using sorted channels under the proposed method. While step-1 and step-3 incur n iterations, time complexity of quick sort based step-2 is O (n log n). Hence, overall time complexity of the proposed TCS calculation methodology is derived as O (2*n + n log n) which is equal to O (n log n).

In comparison to dynamic programming (dp) based problem solving approach which consumed 28 000 iterations, the proposed greedy method in our work incurred only 40 iterations as obtained from the simulation. While the fact that greedy algorithms score better over their dp-based counterpart is already established in literature, the comparison is deliberately performed to highlight the importance of this ratio (28 000/40) with respect to the number of channels. As time complexity is a function of the channel number, this ratio becomes unacceptable with an increasing list of eligible target channels for TCS calculation. The dp-based approach hence can lead to either of the following two consequences – (i) SC computes with limited number of channels (ignoring several other potential channels); (ii) SU includes all eligible channels thereby significantly increasing the number of iterations and leading to loss in time periodicity. The latter one is due to the relationship between the number of algorithmic iterations and the stringent time bound condition for TCS calculation. This is established as follows.

A single SC cycle involves three major operations in our approach as shown in Fig. 2. These include – (1) TCS calculation and distribution; (2) ML-based channel prediction; and (3) SU support during handoff. The first, second and third operations are dependent on the number of available channels, transmitting SUs and SUs with handoff instances respectively. Periodic TCS updates shall occur after completion of every such cycle.

Imperatively, an increase in operation cycle time will reduce the periodic updates of TCS and increase channel obsolescence effects drastically. This imposes time-bound condition on all these operations. With

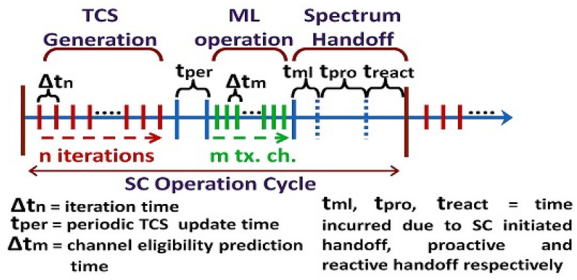


Fig. 2. SC operation cycle with all its time components and related operations.

respect to TCS generation, this implies that the number of iterations n must be kept to a minimum. As already discussed, the dp-based approach fails to collectively maintain both time-bound generation and inclusion of all target channels. On the contrary, the proposed greedy solution with only 0.14% of iterations consumed in dp-based policy addresses both these solutions and is therefore selected in this work.

In addition, comparative evaluation with other greedy approaches is carried out in simulation. As observed from Table 3, the proposed work generates maximum number of target channels (= 40). This implies that the SU can select from a higher number of target channels, thereby reducing call drops during handoff. Secondly, the proposed method records maximum cumulative channel idle time (= 5504.5 ms). This denotes reduced handoff instances by SU due to better selection of target channels with the proposed method.

(iii) Collision Avoidance during Concurrent Channel Access

In order to evaluate channel access conflicts among multiple SUs, the probability of channel selection from proactive TCS must be calculated. In this work, SU senses a channel w_k for n times using energy-detection technique. Therefore, the sensing duration at the jth cycle is given by, $t_s^{(j)} = n \times p$ where p is the duration of one sensing slot. Channel w_k is sensed busy if the received power Pow_{rec} is greater than the threshold power Pow_{th}. The threshold and detected energy levels are given by (11) and (12) respectively.

$$e_{th} = Pow_{th} \times p \tag{11}$$

$$e_{rec} = Pow_{rec} \times p \tag{12}$$

Let us consider a binary variable H_j whose value at jth cycle is set to 1 if $e_{rec} \geq e_{th}$. The channel is rendered busy when $\sum_{j=1}^{n} H_j \geq H_{th}$ where H_{th} is threshold value. Selection of H_{th} is very important. While a higher value increases probability of miss-detection, a lower value will raise false alarm probability.

The probability that channel w_k is sensed busy is given by,

$$\Pr_{busy}^{(w_k)} = \Pr\left[\sum_{j=1}^{n} H_j \geq H_{th}\right] \tag{13}$$

Therefore, the probability that an SU selects a channel w_k from its proactive TCS is expressed as,

$$\Pr_{Pro}^{(w_k)} = \left(\prod_{i=1}^{w_{k-1}} \Pr_{busy}^{(w_i)}\right)\left(1 - \Pr_{busy}^{(w_k)}\right), w_i \in TCS_{pro} \tag{14}$$

It is imperative from (14) that every SU selects the first channel from its proactive TCS with the highest probability in comparison to the other channels. As the channels are sorted in decreasing order of their availability, the probability of the first channel being busy (denoted by $\Pr_{1st-ch_busy}$) is the lowest. Even if such a situation arises, the SU has the option to select from the remaining channels in its TCS.

As soon as SU occupies a channel after handoff, it sends *l_pro* message to the SC. SC updates TCS and User Allocation vectors. During subsequent TCS calculation, SC updates the Channel Allocation vector. Thus, an occupied channel will never be included in the generation of TCS for subsequent SUs. Hence, the probability that an occupied channel is included in TCS of another SU (denoted by $\Pr_{TCS_conflict}$) is 0.

Table 3
Evaluation of different greedy methods for TCS generation.

Metrices	Frac_GR_Desc	Frac_GR_Asc	*Proactive TCS*
Total idle time (ms)	1906	5148.5	**5504.5**
Average no. of channels selected	4	8	**8**

Frac_GR_Desc, Frac_GR_Asc = Fractional Knapsack using Greedy Algorithm with Descending value, Ascending weight respectively.

To this end, there are two possible scenarios of concurrent channel access – (1) when two SUs perform handoff at same time with probability $\Pr_{h|o-same_time}$; (2) when two SUs perform handoff at different timestamps with probability $\Pr_{h|o-diff_time}$. Combining these two cases, the probability of concurrent channel access by two SUs $\Pr_{conflict_PRO}$ can be expressed as,

$$\begin{aligned}\Pr_{conflict_PRO} &= \Pr_{h|o-same_time} \times \Pr_{1st-ch_busy} \\ &+ \Pr_{h|o-diff_time} \times \Pr_{TCS_conflict}\end{aligned} \tag{15}$$

As previously derived, $\Pr_{1st-ch_busy}$ has negligible value whereas the value of $\Pr_{TCS_conflict}$ is zero. Hence, the proposed scheme is successful in avoiding conflict during channel access by multiple SUs in the proactive Phase.

4.3. 2nd phase: Reactive TCS generation and allocation phase

Reactive phase is in effect when SU senses all channels in its predetermined TCS busy during handoff. SU then seeks an updated channel list from SC using *l_react* message. In response, SC creates a list of unallocated channels using the Channel Allocation Vector. SC then generates the reactive TCS by calculating a pseudo-random sequence from this list. SC communicates this TCS to the requesting SU via CCC.

SU randomly selects any one channel from this list, senses the channel and immediately resumes communication if the sensing outcome is idle. Otherwise if the channel is sensed busy, SU selects another random channel from reactive TCS. This procedure may continue till the handoff delay crosses the maximum threshold D_{max} where $D_{max} = D + 1$ s. If SU becomes unsuccessful in selecting an idle channel within D_{max} period, the call is dropped as time-out occurs after this period.

SC is preferred over SU for reactive TCS generation because of the following two reasons.

- Requesting the SC node for reactive TCS reduces channel search time and increases the probability of successfully finding an idle channel. On the contrary, random channel search by SU in absence of any reactive TCS leads to increased call time-outs due to channel unavailability.
- Pseudo-random sequence as generated by the SC node prevents channel access conflict when two SUs with separate session IDs perform handoff at the same time.

Allocation of this reactive TCS introduces additional challenges which are effectively addressed in the following.

4.3.1. Allocation of reactive TCS

Similar to the scenarios under proactive TCS Allocation, reactive TCS may be allocated under the following scenarios.

(i) Scenario 1: *SU handoff in the same PU zone*

This scenario is the same as Scenario 1 discussed under proactive TCS allocation. When an SU (SU_1) selects a channel Ch_1 from its reactive TCS, it notifies SC of its new operating channel with *l_pro* message. Meanwhile, SC on receiving *l_react* message from the other SU (SU_2) of same session ID waits for *l_pro* message from SU_1. As soon as SC receives this message, it forwards channel information to SU_2. Both SUs then exchange RTS-CTS messages and resume their session.

(ii) Scenario 2: *SU handoff in different PU zones*

Under this scenario, one SU performs spectrum handoff to a target channel from its reactive TCS. SC is informed of the new channel via *l_pro* message. The other SU_2 waits for t_{RTS} time for RTP/RTCP message and shifts to CCC on receiving none. SC informs SU_2 of the new operating channel. Both SUs perform RTS-CTS message exchange and resume VoIP communication. It is to be noted that this message exchange procedure among SU_1, SU_2 and SC is the same as explained in Scenario 2 under proactive TCS Allocation section.

4.3.2. Analysis of reactive TCS

Reactive TCS is further evaluated with respect to fair channel allocation and channel access conflict among multiple SUs based on the properties as stated in Table 1.

(i) Fair Channel Allocation

Let n_{react} be the maximum number of channels that can be sensed by SU from reactive TCS. It is expressed as follows.

$$n_{react} = \frac{D_{\max} - (\Delta t_1 + \Delta t_2)}{t_{sw} + t_c + t_s} \tag{16}$$

where, Δt_1 = time incurred by SU in sensing all channels in proactive TCS, Δt_2 = time consumed in the receipt of reactive TCS, t_{sw}= channel switching time, t_c = consensus time, t_s = target channel sensing time. The corresponding list of target channels W_{REACT} is given by,

$$W_{REACT} = \{w_1'', w_2'', \ldots, w_i''\}, \text{ where } |W_{REACT}| = n_{react} \tag{17}$$

Selecting a channel uniformly and randomly from W_{REACT} depends on the channel busy probabilities of all other channels. The probability of selecting a target channel w_k during reactive phase is therefore given by (20). It must be noted that the expression for $\Pr_{busy}^{(w_k)}$ is already derived in (13).

Issue of fair channel allocation arises when multiple SUs from different sessions request SC for reactive TCS. This fairness issue is successfully addressed by the SC node in this work. Firstly, $\frac{1}{|A|+1}$ in (20) ensures that no priority is attached to any particular channel for any user. Secondly, pseudo-random channel sequence guarantees that W_{REACT} will never be same for any two requests. Finally, requests for reactive TCS from different SUs are addressed on a First Come First Serve (FCFS) policy with no priority attached to any certain SU.

(ii) Collision Avoidance during Concurrent Channel Access

Channel access conflict among multiple SUs during reactive phase may occur only with a negligible probability. This is because firstly, reactive phase occurs only in the worst-case scenario in case of failure of the proactive TCS. Secondly, SC based reactive TCS allocation further reduces the chance of collision during channel access as previously described. Mathematically, it can be proved in the following.

- *Proof of negligible channel access conflict in reactive phase*

There are two possible cases of channel access conflict as discussed in the following.

a. Probability that reactive phase occurs: It is the probability that all channels in TCS_{pro} are busy and is given by,

$$\Pr_{TCS_{PRO}_busy} = \prod_{\forall w \in TCS_{PRO}} \Pr_{busy}^{(w)} \tag{18}$$

It is clearly noted that the value of $\Pr_{TCS_{PRO}_busy}$ is negligible with an increasing number of channels in TCS_{pro}.

b. Probability that channel access conflict occurs: This gives the probability with which the same channel from W_{REACT} is accessed by m SUs. It can be expressed in (19).

$$\Pr_{same_ch} = \Pr_{W_{REACT}_same} \times \frac{1}{n_{react}^m} \tag{19}$$

where, $\Pr_{W_{REACT}_same}$ denotes the probability with which the same W_{REACT} is accessed by all SUs.

As can be seen from (19), $\Pr_{same_ch}$ becomes negligible with higher values of m. Moreover, due to pseudo-random sequence in W_{REACT}, value of $\Pr_{W_{REACT}_same}$ is also minimal. See the Eqs. (20) and (21) given in Box I.

Combining these two cases, the probability of channel access conflict during the reactive phase is given by (21). Negligible values in (18) and (19) imply that the value of $\Pr_{conflict_REACT}$ is also negligible. This guarantees collision avoidance during channel access under the reactive phase.

4.4. 3rd phase: Periodic TCS generation and allocation phase

This phase is in effect during SU transmission when there is no handoff operation. SC calculates channel conditions at regular intervals. Based on sensing outcome, SC periodically generates TCS and allocates this periodic TCS to the participating SUs within a VoIP session. The TCS generation method is the same as followed in proactive TCS calculation. The subsequent allocation process is discussed in the following.

4.4.1. Allocation of periodic TCS

During normal SU transmissions, periodic TCS is broadcast by the SC node to the subscribed SUs. A special scenario arises during the initiation of a VoIP call, as explained below.

Let SU_1 and SU_2 be the caller and callee SUs respectively. SU_1 requests for a default channel from SC. It receives channel information via the Common Control Channel (CCC) and initiates the call. Based on the session ID in User Allocation Vector, SC broadcasts this channel message so that SU_2 listening on another idle channel can receive it. Finally, SU_1 and SU_2 exchange RTS-CTS messages, initiate call signaling process and start the VoIP call.

It is worth mentioning that allocation of CCC and default channel are integral toward execution of the proposed methodology. Notable studies in literature such as [6,11,14,28,31,44,52] have relied on CCC during handoff. However, CCC itself may turn obsolescent due to PU activities, resulting in inefficient handoff operation.

This requires dynamic selection of CCC based on channel conditions as followed in some studies [15,26]. These works have sorted target channels in either ascending order of channel activities [26] or in descending order of packet error rates [15]. The first channel in the sorted list is marked as CCC. However, these studies considered a distributed network without much emphasis on CCC allocation and its updates.

On the contrary, the periodic phase in this paper ensures CCC allocation as well as its regular updates in a centralized network. This is further discussed in the following.

4.4.2. Selection of CCC

During periodic TCS generation, SC selects the first channel from the sorted channel list as CCC. This ensures that the channel with the highest ratio of effective to non-effective communication period is selected as a reliable medium for interaction. SC then updates its Channel Allocation Vector and broadcasts this CCC information with periodic or proactive TCS (depending on their occurrence) to all the SUs.

Periodic TCS generation also ensures that the CCC is updated regularly. This prevents unavailability of CCC under channel obsolescence effects. In the worst case, it may so happen that CCC becomes unavailable even before periodic TCS updates are made. SC addresses this scenario by deploying one of its two transceivers to monitor the condition of CCC when there is no requirement for communication with SUs. If CCC is sensed busy, SC immediately selects the new CCC as another channel that satisfies the following criteria.

- Channel value in the Channel Allocation Vector is not 1.

$$\underset{React}{\overset{(w_k)}{\Pr}} = \left(1 - \underset{busy}{\overset{(w_k)}{\Pr}}\right) \sum_{\forall A \subseteq W_{REACT}/\{w_k\}} \left[\frac{1}{|A|+1} \prod_{\forall v \in A} \left(1 - \underset{busy}{\overset{(v)}{\Pr}}\right) \prod_{\forall v' \notin A, \forall v' \notin \{w_k\}} \underset{busy}{\overset{(v')}{\Pr}}\right], w_k \in W_{REACT} \tag{20}$$

$$\underset{conflict_REACT}{\Pr} = \underset{h|o-same_time}{\Pr} \times \prod_{\forall w \in TCS_{PRO}} \underset{busy}{\overset{(w)}{\Pr}} \times \underset{W_{REACT_same}}{\Pr} \times \frac{1}{n^m_{react}} \tag{21}$$

Box I.

- Channel has the highest ratio of effective to non-effective communication period in the present sorted channel list.

Finally, the new CCC information is broadcast to all SUs and the corresponding vectors are updated.

4.4.3. Allocation of default channel

A newly admitted SU must also be provided with a default channel. An idea consistent with the proposed scheme is that the SU sequentially senses all channels (starting from the lowest frequency). After finding an idle channel, it waits for periodic CCC broadcast from the SC. On receiving CCC information, SU uses this CCC to transmit channel request message to the SC. SC selects one idle channel by consulting its Channel Allocation Vector and allocates it to the SU. On successful allocation, SC registers SU and updates its vectors. Notably, selection of default channel depends on call admission control policies and is beyond the scope of discussion in this paper.

It is thus inferred that the proposed scheme in three phases provides a comprehensive solution to TCS generation and allocation problem for multiple SUs under dynamic channel conditions. This is further enhanced by the inclusion of ML-enabled CSP block in SC and is discussed in the next section.

5. Proposed HMM-BiLSTM framework

A major concern from the previous section is that the reactive handoff phase may underperform when several SUs experience similar failure in their proactive TCS. This may, in turn, lead to call drop as the affected SU may not be able to resume call in the new target channel within the D_{max} threshold. To solve this problem, two solutions are envisaged as follows.

(i) Reduce the probability of reactive handoff by SU by initiating spectrum handoff from the SC end;
(ii) Reduce the network control overhead in the CCC so that even if reactive handoff occurs in the exceptional scenario, the affected SU may instantly transmit *l_react* message to SC.

While the second solution is addressed in Section 6.6, this section implements the first solution through incorporation of an intelligent CSP (Channel State Prediction) block. This block examines every active channel under transmission from *C_V* vector and predicts channel status using a novel machine-learning model. If the predicted outcome suggests degraded channel state, SC initiates spectrum handoff on behalf of SU. This is achieved by transmitting *ch_react* message to SU that intimates SU to immediately vacate the current channel. SU selects the next target channel from its TCS, transmits *l_pro* message and receives updated periodic TCS from SC. Thus the probability that proactive/periodic TCS may outperform leading to reactive handoff is drastically reduced by this CSP block. Specifically, this paper proposes a novel HMM -BiLSTM framework that learns and predicts VoIP specific channel eligibility status.

The HMM model [80] generates the optimal sequence of channel states after observing call quality for each state. This is further fed to the BiLSTM model [81] that performs sequence classification and finally predicts the channel conditions. The designed model along with its parameters is illustrated in Fig. 3 and is explained as follows.

5.1. HMM component of the CSP block

A novel 2-order HMM is designed using both first and second order metrics that shall predict the optimal sequence of states for a particular channel. HMM is selected because the actual channel conditions for suitably hosting VoIP communication are hidden and can only be derived using channel statistics recorded using different monitoring tools in the test-bed. Specifically, HMM is advantageous in two ways. – (1) inter-sensing duration T_{SI} used by SC during channel selection is prone to miss-detection and is avoided by HMM; and (2) calculated VoIP transmission time (T_v) varies with respect to its actual value (T_{su}). HMM avoids this discrepancy in determining VoIP transmission slot.

The first order metric is the mean channel off period (μ_{off}) as recorded by SC over a duration of 20 min. The second order metric denotes variation in channel idle period. This is crucial because a channel with high mean off period may not always be VoIP friendly if it frequently toggles between busy and idle states. This variation as denoted by var_{off} is derived as follows.

For a particular channel, the off period is recorded in a 1-D matrix with *A* symbols. The co-occurrence matrix *AXA* is defined such that an element $a_{i,j}$ of this matrix contains the ratio of the sum of the number of contiguous data pairs with values of *i* and *j* to the sum of the total number of all possible pairs. This is explained using an example. Let *A* contain the mean off period values (in ms) as follows : [30, 280, 280, 110, 30, 280, 280, 250, 100, 90]. Based on these values, the co-occurrence matrix of size 10 × 10 is created where the rows and columns are evenly spread out between 0 and 280. $a_{10,10}$ corresponds to the contiguous data pair (280, 280). Let us consider that this pair is repeated twice within a space of total 9 pairs. Therefore, we have $a_{10,10}$ = 2/9. Similarly, other values of $a_{i,j}$ are filled up accordingly. Once this *AXA* matrix is generated, var_{off} is calculated as follows.

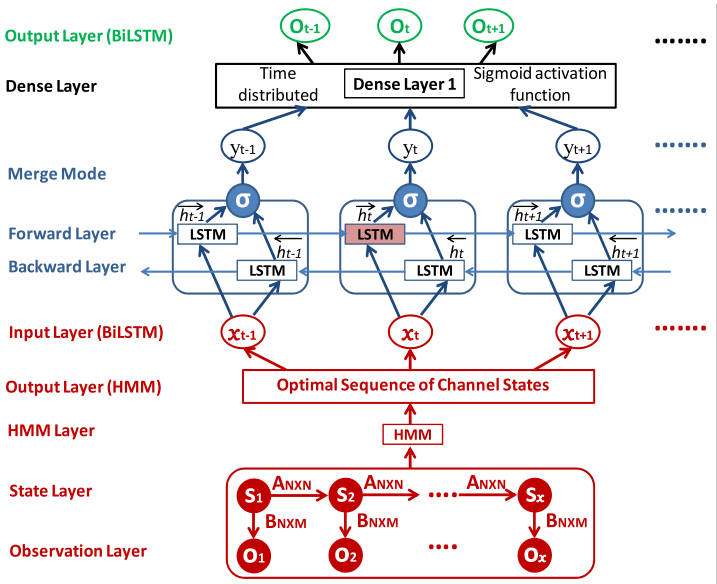
$$var_{off} = \sum_{i,j} (j-i)\,|j-i|\,a_{ij} \tag{22}$$

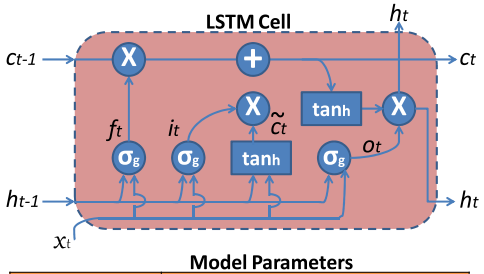
Based on the range of μ_{off} and var_{off}, the number of hidden states is defined. For example, state 1 corresponds to the hidden state for which μ_{off} is less than 50 ms and var_{off} is −20. A total of 20 such states are defined such that the *i*th state is better than $(i-1)$th state in terms of call quality. Accordingly, the proposed HMM model in this paper is characterized by $\{Ns, s, Mo, O, A_{NsXNs}, B_{NsXMo}, \pi\}$ where these are defined as follows.

(a) *Ns* denotes the total number of states in the model where the states are denoted by $s = \{s_1, s_2, \ldots, s_{Ns}\}$.
(b) *s* denotes an individual state. Formally, for a particular channel w_i, a state s_i is defined as follows.

$$s_i = \{x, y\} \;\; where\; x \in \mu^i_{off}, y \in var^i_{off} \tag{23}$$

(c) *Mo* denotes the total number of observation symbols where the symbols are denoted by $O = \{o_1, o_2, \ldots, o_{Mo}\}$.
(d) *O* denotes the observation symbol that corresponds to VoIP call quality in terms of MOS (Mean Opinion Score) under that particular channel condition and is provided by the test-bed recordings. Formally for a particular state s_i, the observation





Model Parameters

HMM		BiLSTM			
Parameter Type	Metric used	Parameter Type	Metric used	Parameter Type	Metric used
No. of states	20	Model type	Sequential	No. of neurons	20
No. of observation symbols	16	Dense layer type	Time distributed	No. of time-steps	10
No. of parameters per state	2	Model return type	Return-seq. based	No. of features	1
Transition prob. matrix	20x20	Activation mode	Sigmoid	No. of epochs	250 to 1000
Observation prob. matrix	20x16	Learning parameter optimization	ADAM's optimizer	No. of dense layers	1
Optimal sequence no.	10	Merge mode	Sum		

Fig. 3. Proposed HMM-BiLSTM framework along with detailed view of each LSTM cell and complete description of all the model parameters.

symbol o_s is given by the MOS value for channel w_i under the channel conditions defined by state s_i.

(e) $A_{N_S X N_S}$ denotes the state transition probability matrix.

(f) $B_{N_S X M_O}$ denotes the observation probability matrix and is defined as follows.

$$B_{N\times M} = \{b_{jk}\} \ \ where\ b_{jk} = \Pr\left(o_k \ at\ time\ t \mid S_t = s_j\right), \quad 1 \le j \le N,\ 1 \le k \le M,\ t = 1, 2, 3, \ldots \tag{24}$$

(g) π denotes the initial state distribution vector and is defined as,

$$\pi = \{\pi_i\} \ \ where\ \pi_i = P\{S_1 = s_i\}, \quad 1 \le i \le N_s \tag{25}$$

Once the model is established, the model parameters are further estimated using the E–M (Expectation–Maximization) algorithm that relies on maximum likelihood estimate [17]. Using the Viterbi algorithm [17], the optimal state sequence is searched using dynamic programming methodology. Formally, the optimal channel state sequence $S_{optimal}$ for a particular channel w_i and the associated observation O is the one having the maximized probability $Pr(S|O,\lambda)$ where $\lambda = \{A_{N_S X N_S}, B_{N_S X M_O}, \pi\}$.

Similarly for other channels in the network, the corresponding optimal sequences are obtained using this procedure and are finally fed to the BiLSTM model which is described as under.

5.2. BiLSTM Component of the CSP block

The objective is to take as input the optimal sequence of channel states (generated by HMM) and classify the underlying channel as either eligible for VoIP calls or not. BiLSTM is chosen for this purpose and its selection is justified as follows.

It is well explored in literature that traditional neural networks are incapable of maintaining temporal dependencies in input data sequences. On the contrary, both RNN and LSTM succeed in capturing the temporal relationships using memory cells where the output derived from some layers are fed back using a feedback loop [81]. However, LSTM cell using its input, output and forget gates (as shown in Fig. 3) scores over RNN in terms of remembering long-term dependencies as well as tackling the vanishing gradient problem [81]. Additionally, forward and backward LSTM layers are combined in bidirectional LSTM (or BiLSTM) that tackles both forward as well as backward dependencies in data sequences. In essence, there are two hidden layers (instead of one) in BiLSTM that connect to the output layer. BiLSTM, therefore, is best suited to the given problem in this paper namely, channel state classification based on obtained sequence. This is because the obtained channel sequence exhibits both forward as well as backward long-term temporal dependencies. The forward relationship is attributed to PU activity that drives the channel states depending on its on–off behavior. VoIP activity, on the contrary, can lead to backward dependencies among channel states. This is because the current channel context may be derived using the sender and receiver reports generated in near future by the VoIP call signaling protocol (SIP or H.323 [78]). The same is validated by recording superior performance of BiLSTM in Section 6.

The model parameters for BiLSTM are expressed as follows.

$$f_t = \sigma_g\left(z_f x_t + y_f h_{t-1} + b_f\right) \tag{26}$$

$$i_t = \sigma_g\left(z_i x_t + y_i h_{t-1} + b_i\right) \tag{27}$$

$$q_t = \sigma_g\left(z_q x_t + y_q h_{t-1} + b_q\right) \tag{28}$$

$$\tilde{C}_t = \tanh\left(z_c x_t + y_c h_{t-1} + b_c\right) \tag{29}$$

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \tag{30}$$

$$h_t = q_t \times \tanh\left[C_t\right] \tag{31}$$

$$L_t = M\left(\vec{h}_t, \overleftarrow{h}_t\right) \tag{32}$$

where z_f, z_i, z_q, z_c = weight matrices from hidden input layer to three gates and input cell state; y_f, y_i, y_q, y_c = weight matrices from previous cell output; b_f, b_i, b_q, b_c = bias vectors; σ_g = sigmoid function; $tanh$ = hyperbolic tangent function; C_t = cell output state; h_t = layer output; L_t = output from BiLSTM; M () = merging function of 2 LSTMs.

The selection of model parameters as per Fig. 3 is justified as follows. Firstly, sequential model type enables one to quickly construct BiLSTM by adding layers in the sequence in which they need to be operated. This greatly reduces the model building time. Secondly, use of a fully connected dense layer with time distributed wrapper aids in predicting one output per timestamp as denoted by the uppermost layer in Fig. 3. This, in turn, requires LSTM hidden layers to return sequence rather than single value. Hence, model return type is return-sequence based. Moreover, sigmoid activation function is selected as this is a binary classification problem (channel is classified as eligible or not). Model compilation is facilitated by a recently designed research

technique — Adam's optimizer [81] that derives weights using adaptive moment estimation. This records significant gains in training speed (lesser time complexity). Finally, the selection of "sum" function as the merge mode is primarily data-centric and its superiority over other functions is realized in Section 6.5.

Notably, CSP block equipped with the HMM-BiLSTM model is operated in parallel with other SC operations such as TCS calculation and SU coordination operations. Since this block operates only after successful handoff, the time complexity of this model does not affect the stringent spectrum handoff delay requirements of VoIP calls. Rather, this model aims to keep channel prediction time to a minimum for quick decision-making by SC and is suitably demonstrated in the following section.

6. Performance evaluation

This section establishes the importance of the key design features in this work with respect to the following scenarios.

(i) Multi-Channel scenario — It reflects the adaptive channel selection process under dynamic network conditions.
(ii) Multi-User scenario — Primary focus lies on fairness and collision avoidance during channel access.
(iii) Single-User scenario — It evaluates the handoff performance for SUs engaged in VoIP communication.
(iv) ML scenario — It evaluates the enhancement achieved through incorporation of the CSP block.
(v) Network Control overhead — It studies the impact of overhead imposed by the control packets and establishes both the necessity and advantages of the proposed approach.

6.1. Simulation model

The simulation setup is developed in MATLAB. Sensing and transmission slots are simulated using non-integral multiples of 10 ms slot based on IEEE 802.22 standard [76]. The timing parameters are selected as per VoIP traffic requirements. These include – (1) sensing time t_s = 100 ms; (2) transmission time t_d = 250 ms; (3) t_c = 100 ms; (4) t_{sw} = 25 ms; and (5) D_{max} = 2 s.

Channel conditions are modeled using exponential on–off PU traffic [77]. The network is rendered dynamic by uniformly varying the channel transition rates α, β. These rates are based on three activity factors 0.2, 0.3 and 0.5 denoting low, medium and high PU traffic respectively. Table 4 provides details of these metrics. For example, an activity factor of 0.2 denotes 20% on-time and 80% off-time. Each activity factor is assigned a probability value based on which four scenarios resembling dynamic channel conditions are generated. As an instance, Scenario I varies the low, medium and high activity factor values with probabilities 0.7, 0.2 and 0.1 respectively.

Without loss of generality, the following five techniques are used for comparative study in the simulation model.

(i) *React*: This refers to random on-demand selection of the target channels by the SU after performing handoff [35].
(ii) *Pro_idle*: This refers to the proactive selection of target channels based on their idle periods, as followed in [25].
(iii) *Pro_TI*: This denotes proactive TCS calculation based on the values of PU Traffic Intensity (TI) [23].

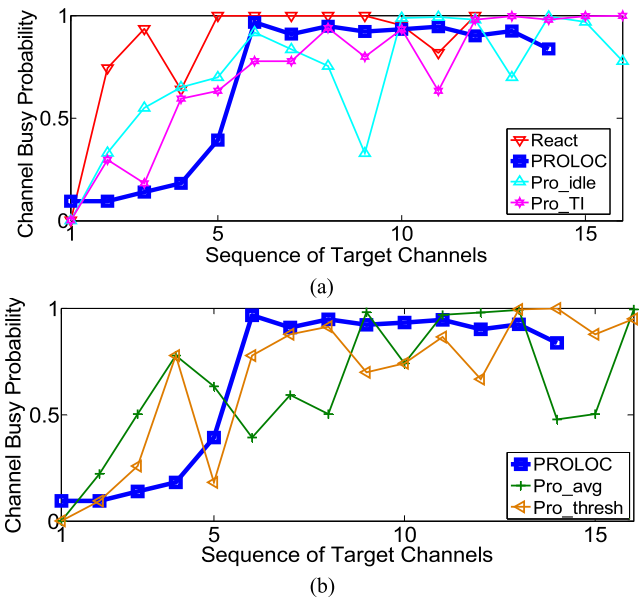


Fig. 4. Channel busy probabilities of target channels under different schemes.

(iv) *Pro_avg*: Here SUs calculate time-average idle probability of channels based on remaining OFF periods [5,30].
(v) *Pro_thresh*: Here only those target channels are selected whose idle period probability is greater than a threshold value for successful frame transmission [21,28,30].

This paper lays more emphasis on the underlying techniques rather than the specific methods used in such works. The proposed method in our paper is denoted by PROLOC which signifies Proactive and Reactive TCS, Optimal channel allocation, Low complexity and Collision Avoidance.

6.2. Multi-channel scenario

Fig. 4 demonstrates the sequence of target channels selected by all techniques under the dynamic channel conditions as per Table 4. Evaluation is done on the following three parameters.

(i) Average target channel busy probability – A low value implies reduced connection disruption rate;
(ii) Total number of selected channels –A high value denotes reduced probability of call drop;
(iii) Nature of the curve for channel busy probability — More spikes imply random channel section. This, in turn, records more handoff instances and call drops.

As observed from Fig. 4(a), *React* performs on-demand sensing and selection resulting in channel busy probability of 1. Moreover, random spikes are recorded for *Pro_idle* and *Pro_TI*. This is because they use well-defined traffic models for TCS generation and thus suffer from channel obsolescence effects. In addition, *React*, *Pro_idle* and *Pro_TI* record hikes in channel busy probability from the 2nd, 3rd and 4th channel respectively. Comparatively, PROLOC selects target channels

Table 4
Modeling of dynamic channel conditions.

	Activity factor	Transition rates		Time periods (ms)		Probabilities of scenarios			
		α	β	t_{on}	t_{off}	I	II	III	IV
Low	0.2	0.5	0.125	180	720	0.7	0.5	0.4	0.3
Medium	0.3	0.33	0.142	270	630	0.2	0.4	0.3	0.3
High	0.5	0.2	0.2	450	450	0.1	0.1	0.3	0.4

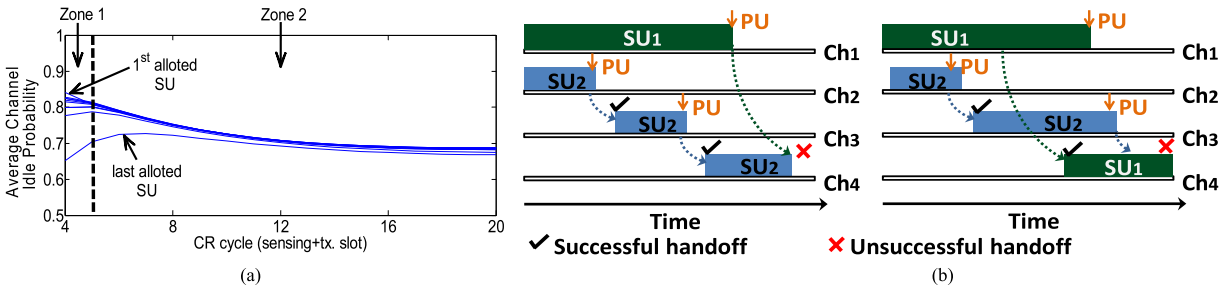


Fig. 5. (a) Fair Channel Allocation in terms of the average idle probability of the selected target channels by multiple SUs as per the proposed scheme; (b) Illustration of collision during channel access by multiple SUs after executing simultaneous spectrum handoff operations.

with lower busy probabilities with channel hike recorded only after the 6th channel. Finally, it records a uniform trend in channel selection by choosing channels in increasing order of busy probabilities. This denotes adaptive channel selection under dynamic conditions and is attributed to periodic TCS calculation by SC.

Fig. 4(b) reveals that *Pro_thresh* and *Pro_Avg* record channel busy probability of 1 due to obsolescence effects under dynamic conditions as they perform short-term prediction of channel usage statistics. Conversely, reactive TCS in PROLOC maintains channel busy probabilities below 0.8. Thus, it restricts the amount of channels under D_{max} constraint. On the contrary, proactive TCS ensures target channel selection with lower busy probabilities as compared to *Pro_thresh* and *Pro_avg*.

It is thus inferred that the proactive, reactive and periodic phases cooperate toward adaptively generating the TCS sets.

6.3. Multi-user scenario

This section considers two criteria for evaluation – (1) fair channel allocation; (2) collision avoidance during channel access.

6.3.1. Fair channel allocation

An unfair scenario is one where one category of SUs is allocated channels with higher idle probabilities in comparison to the others. This issue is studied in this paper using the time-average idle probability of the target channels as the fairness metric. Selection of this metric is justified in the following.

(i) Selection of Fairness Metric

Notable studies in literature [5,24] have chosen fairness metrics that are not applicable in this work. Firstly, uniform handoff delay metric [5] may not clearly reflect the number of handoff operations for an SU. For example, a TCS having channels with low time-average channel idle probability will witness higher number of handoff instances in comparison to a TCS with high time-average channel idle probability.

Secondly, MOS [24] is an application-level QoS metric. Therefore, it is not clear whether degradation in MOS is due to TCS related issues or other factors involved in communication such as echo, encrypted communication, equipment impairment etc. Hence, this metric cannot be used as the absolute metric to study fair channel allocation in this paper.

Conversely, the selected metric in this paper provides a clear view of fair channel allocation as discussed in the following.

(ii) *Application of Fairness Metric*

Fig. 5(a) records the average idle probability of the allocated target channels to different SUs. In an unfair scenario, SUs with channels of higher idle probabilities would have recorded higher curves than the others, resulting in unevenly spaced curves. On the contrary, the closely spaced curves in Fig. 5(a) signify fair channel allocation.

In a notable observation, Zone 1 in Fig. 5(a) denotes a noticeable gap in the beginning of the 4th time slot. This is because after periodic TCS updates, the best target channel is allotted to the first SU, followed by allocation of the second best channel to the next SU and so on in a FCFS (First Come First Serve) policy. Thus channels allotted to successive SUs will have lower idle probabilities compared to the previous SUs. However, with increase in time slots, these probabilities become nearly identical in Zone 2 indicating fair treatment to all SUs.

6.3.2. Collision avoidance during channel access

It is also evident from Fig. 5(a) that all curves follow a uniform pattern without any sharp decline along the x-axis. This ensures that handoff by one SU remains unaffected by other SUs. This is explained by an illustration in Fig. 5(b).

Let SU_1 and SU_2 be equipped with TCS $\{Ch_4, Ch_7, Ch_5\}$ and $\{Ch_3, Ch_4, Ch_6\}$ respectively. SU_1 and SU_2 are operating in Ch_1 and Ch_2 respectively. In absence of collision avoidance scheme, two scenarios may arise as explained in the following.

- Scenario 1: SU_2 performs multiple handoff operations and moves to Ch_4. When SU_1 tries to access Ch_4 during handoff, it finds Ch_4 busy although Ch_4 is enlisted in its TCS.
- Scenario 2: SU_1 switches to Ch_4. SU_2 attempts to access Ch_4 after its second handoff operation. However, Ch_4 is detected busy as it is occupied by SU_1.

Both these scenarios are demonstrated in Fig. 5(b). It is evident that the handoff operation of one SU is affected by another SU. Due to this unexpected unavailability of idle channel, the average idle probability of the selected target channels will experience sharp decline in those time slots where such collisions occur. On the contrary, uniform curves in Fig. 5(a) ensure that there is no such conflict among SUs.

6.4. Single-user scenario

This scenario analyzes QoS degradation during spectrum handoff under three performance metrics – (1) handoff delay; (2) successful transmission period; and (3) call drop probability.

Let $N = n_{pro} + n_{react}$ be the number of channels sensed by an SU in proactive and reactive TCS respectively. Under the condition of successful handoff, the probability that handoff occurs after sensing k channels is given by,

$$\Pr_{h|o} = \Pr\left(h = k / h \leq N\right) = \frac{\Pr\left(h = k \bigcap h \leq N\right)}{\Pr\left(h \leq N\right)} \tag{33}$$

where, h is a temporary variable indicating that SU performs handoff after sensing $h = k$ number of channels. As handoff may occur either from the proactive/periodic TCS or reactive TCS, the following two cases arise from (33).

- Case 1: Probability that idle channel is selected at kth trial from proactive/periodic TCS — This has been derived in (14).
- Case 2: Probability that idle channel w_k is selected at kth trial in reactive TCS : $\Pr_{React}^{k^{th}\ trial}$ – This is derived as follows.

$$\Pr_{React}^{k^{th}\ trial} = \prod_{l=1}^{n_{pro}} \Pr_{busy}^{(w_l)} \times \prod_{l'=n_{pro}+1}^{k-1} \Pr_{busy}^{(w_{l'})} \times \left(1 - \Pr_{busy}^{(w_k)}\right) \tag{34}$$

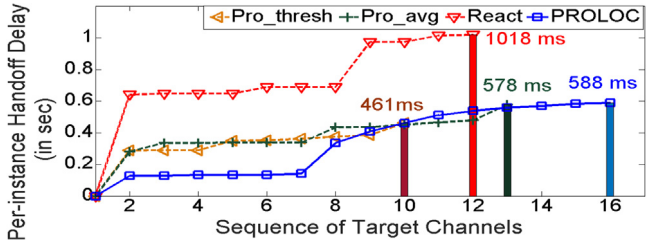


Fig. 6. Average per-instance handoff delay for selecting each target channel from TCS under different policies; Mean delay is recorded.

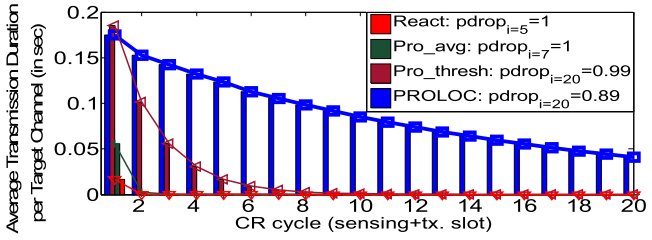


Fig. 7. Average successful transmission duration under different policies.

Table 5
Call drop probability for the different schemes.

	React	Pro_avg	Pro_thresh	PROLOC
CR cycle when call drops: m	5	7	20	**20**
Pr^{m}_{drop}	1	1	0.99	**0.89**

The mean handoff delay is given by the product of the expected number of channels sensed before handoff and the different timing parameters. It is therefore expressed as,

$$d_{h|o} = \left[\sum_{k=1}^{N}\left[k \times \Pr\left(h = k \Big/ h \le N\right)\right] \times t_s\right] + t_c + t_{sw} \tag{35}$$

Fig. 6 plots this handoff delay $d_{h|o}$. *Pro_idle* and *Pro_TI* are not considered due to their degraded performance as per Fig. 4(a). The highest delay is observed for *React* due to random channel selection. In comparison, *Pro_thresh* and *Pro_avg* record lower delays due to efficient channel selection. PROLOC records superior performance with respect to all these methods. It adaptively ensures that $d_{h|o}$ is below 150 ms during proactive/periodic phase (40% lower than *Pro_thresh*, *Pro_avg*) and 600 ms in reactive phase (60% lower than *React*).

The reason for this reduced delay is that PROLOC performs low-complexity TCS calculation and allocation under D and D_{max} threshold limits. Thus it succeeds in keeping $d_{h|o}$ under 588 ms. This is further justified for VoIP applications because of the reasons stated as under.

- End-to-end delay threshold for VoIP ranges within 200 ms. Hence, recorded delay will not significantly degrade QoS;
- Call time-out duration is set to a minimum 1 s (default 60 s) under CISCO VoIP specifications [36]. Hence, $d_{h|o}$ value under 588 ms in PROLOC ensures successful resumption of VoIP call after handoff without any time-out.

Better channel selection also ensures the highest transmission duration per target channel for PROLOC as observed in Fig. 7. This implies lesser handoff instances and reduced QoS degradation for VoIP SUs. However as CR cycle progresses, the idle probability of the selected target channel decreases. As a result, all methods witness reduction in their transmission durations. This also increases their call drop probabilities.

Let call drop probability at mth cycle be denoted by Pr^{m}_{drop}. It is recorded for different methods in Table 5. *React*, *Pro_avg* and *Pro_thresh* record maximum values with *React* and *Pro_Avg* witnessing call drops as early as the 5th and 7th cycle respectively. This is attributed to their handoff delays which result in early call time-outs. Conversely, PROLOC records the lowest call drop value owing to its least handoff delay under proactive and reactive phases. In another significant contribution, PROLOC is successful in sustaining VoIP calls even after 20th cycle preventing call-timeouts under dynamic channel conditions.

It is thus inferred that PROLOC fulfills the criteria of fairness and collision avoidance while sustaining QoS aware VoIP calls.

Table 6
Accuracy vs. Model building time for different models.

	No. of epochs = 250		No. of epochs = 1000	
	Time (s)	Accuracy	Time (s)	Accuracy
Forward LSTM (lstm_forw)	21.1	0.91	47.1	0.93
Backward LSTM (lstm_back)	21.2	0.9	49.9	0.92
BiLSTM (bilstm_sum)	**23.5**	**0.95**	**57.4**	**0.97**

6.5. ML scenario

This section studies SC performance with the integration of CSP block. This enhanced methodology is hereby denoted as PROLOC-ML. Performance evaluation of this ML component is carried out in three phases namely – (1) comparative analysis with other similar ML techniques; (2) parameter configuration and model performance evaluation; and (3) performance gain with respect to the basic PROLOC.

The simulation platform equipped with channel conditions is fed with VoIP metrics using MATLAB Simulink tool. Once transition and observation probability matrices are developed, HMM based on the first and second order metrics is simulated in MATLAB and imported in Python for integration with the BiLSTM model. Binary log loss (binary_cross entropy in Keras) is used to compare accuracy of all LSTM techniques.

Firstly, it is observed in Fig. 8(a) that BiLSTM with merge mode as "sum" (denoted by bilstm_sum in figure) scores better in terms of minimizing log loss values compared to basic LSTM methods. While the logic behind its superior performance is highlighted in Section 5, it is noted that lower loss implies higher accuracy which incurs the cost of rising time complexity. Accuracy is further enhanced by training the model for a larger number of epochs. Table 6 highlights the tradeoff among accuracy, number of training epochs and time incurred in model building for all methods. It is worth mentioning that accuracy metric in Table 6 refers to accurate prediction of channel state under a test sequence given a set of training sequences.

Intuitively, loss in model accuracy shall affect handoff performance of SUs. For instance, in case of false-negative cases (analogous to false alarm), an SU may unnecessarily vacate its operating channel. The situation is more alarming for false-positive cases (similar to miss-detection) where SU continues transmission despite channel conditions being adverse. This greatly increases the probability of reactive handoff and defeats the purpose of CSP block. Accordingly, the objective is to select a highly accurate model (BiLSTM) in this scenario and regulate the number of epochs depending on model building time incurred by the underlying hardware. However, a large number of epochs result in the overfitting problem which must be avoided. To this end, mean square error (mse) metric is used to justify the epoch selection in Fig. 9. It is clearly observed that as the number of epochs increase beyond 250, the accuracy decreases due to overfitting problem. Therefore epoch selection is a critical aspect in model formulation for this work.

It is to be noted that Table 6 reflects relative trends among different models. Their absolute values depend on the dataset and hardware. Another point of interest is that the overall network performance depends more on channel prediction time (which must be in the order of μs) rather than model building/ training time. Both these issues are explored in Section 7.

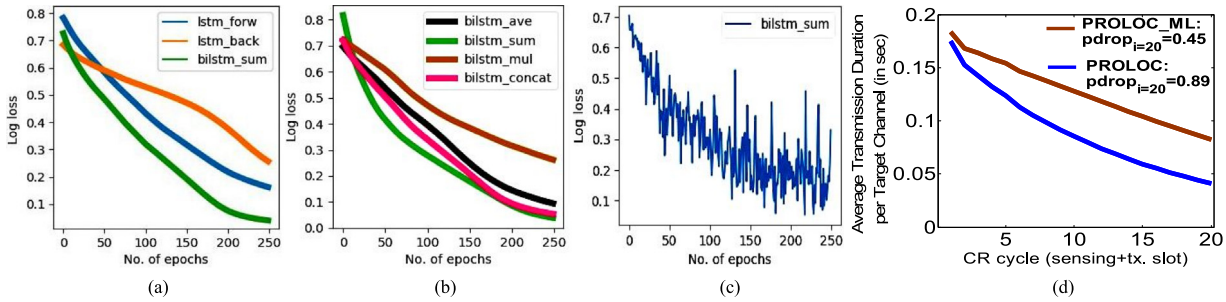


Fig. 8. (a) Comparative analysis of the proposed HMM-BiLSTM model with respect to (a) related LSTM models; (b) different merge mode functions; and (d) PROLOC framework. (c) Evaluation of HMM-BiLSTM model with different training epochs under dynamic channel conditions.

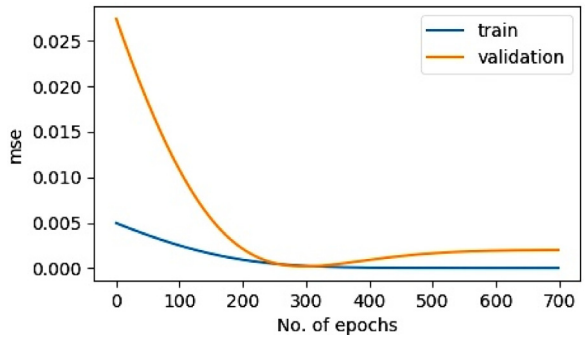


Fig. 9. Evaluation of the overfitting problem with respect to the number of epochs on the model accuracy (mean square error).

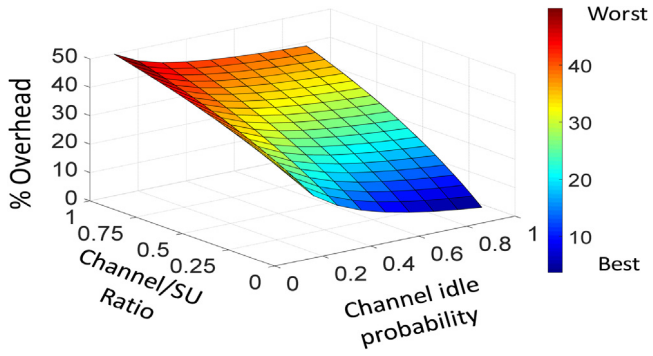


Fig. 10. Percentage of the control packet overhead recorded from SU end.

With focus on increasing accuracy under BiLSTM, the next step is to evaluate the model under log loss values for different merge modes. It is seen in Fig. 8(b) that "sum" mode performs comparatively well in obtaining higher accuracy. In addition, "sum" is based on linear combination and is simpler compared to "mul" or "concat" modes thereby adding to the practical applicability requirement of this work. An interesting observation from Fig. 8(a) and (b) is that the model learns differently every time it is run on the dataset. This accounts for minor variations in bilstm_sum values in these figures.

The BiLSTM model configured with the parameters of Fig. 3 is thereafter studied under different channel sequences. The performance gain in terms of reduced loss in accuracy is shown in Fig. 8(c). However under dynamic channel conditions, the training rate of the model is nonlinear resulting in those peak values. These peaks correspond to the heterogeneous scenario where the model still learns from its dataset that varies abruptly owing to unpredictable behavior of different channels.

Finally, PROLOC_ML is compared with PROLOC. It is seen in Fig. 8(d) that the average transmission duration per target channel increases drastically for PROLOC_ML. This is because in the presence of ML-enabled CSP block, SC alarms every SU of degrading channel state. SC thereby feeds SU with updated periodic TCS every time SU switches to the next target channel. Thus on an average, SU is more updated with channel information for PROLOC_ML, thereby leading to better target channel selection and subsequent increase in call transmission time per channel. Additionally, call drop rate is reduced from 0.89 for PROLOC to 0.45 for PROLOC_ML at the end of the 20th cycle. This is primarily attributed to lesser occurrences of reactive handoff and associated call drops due to RTP/RTCP time-out.

It is thus inferred that PROLOC_ML enhances the performance of PROLOC with better channel prediction and minimizes reactive handoff instances and associated call drops.

6.6. Network control overhead scenario

Finally this segment provides an exhaustive review of network control overhead generated during the entire operation of PROLOC-ML. This is critical to system performance as the overall system throughput must not be disrupted by such control overhead. Fig. 12 summarizes all conditions for generation of control packets. The message notations are stated as follows – m_1 = Periodic TCS; m_2 = *l_pro* message + Proactive TCS; m_3 = *l_react* message + Reactive TCS; m_4 = consensus message; m_5 = *ch_react* message; m_6 = operating channel message. Moreover, CCC information is transmitted as part of periodic or proactive TCS and is not explicitly analyzed. Based on these messages, four cases may occur as detailed below.

(i) Normal SU transmission: Periodic TCS (m_1) is transmitted across the operating channel.
(ii) SU proactive handoff: This is either initiated by SU or SC. SU-initiated handoff consumes m_2 and m_4 message, whereas SC-based handoff incurs m_5, m_2 and m_4 message.
(iii) SU-based reactive handoff: This is performed by SU when both periodic TCS and ML methods underperform. It consumes m_3 and m_4 messages. Unlike the previous two cases, this operation occurs entirely on the CCC.
(iv) SU-based forced handoff: This scenario occurs rarely when one SU has performed handoff, forcing the other to move to CCC and receive the operating channel from SC.

Based on these cases, analysis is performed in simulation with respect to three metrics – (1) SU end: average control message overhead; (2) SC end: overall time utilization by control packets; and (3) Channel end: Free spectrum utilization by control packets (Operating channel and Common Control channel). Test-bed observation in Section 7 further validates this analysis.

Firstly from SU perspective, the control message overhead is plotted in Fig. 10 with respect to mean channel idle probability and ratio of number of channels to the number of active SUs. In the worst case, the channel/SU ratio as well as the channel busy probability is 1. This implies that no idle channel is left in the system leading to call drops. Such a scenario consumes 50% of the total messages reported from

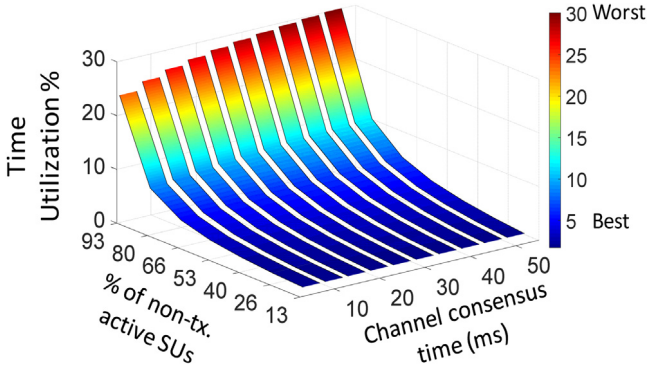


Fig. 11. Time utilization overhead by control packets in the network.

SU end. For the average case corresponding to channel/SU ratio of 0.5 and channel idle probability of 0.5, not more than 30% of the total messages are occupied by network control packets. This confirms minimal degradation to SU transmission by network control overhead.

Secondly from the SC end, it is shown in Fig. 11 that these control messages incur maximum 30% of the total time. The primary factors behind this consumption are the following.

(i) Active SUs that are in handoff mode are chief contributors to the network control overhead. This is evident in Fig. 11 and is justified in Fig. 12. This is because all scenarios except the leftmost case in Fig. 12 generate control messages for non-transmitting active SUs.
(ii) Channel consensus messages are involved for all such SUs involved in handoff as depicted in Fig. 12. Henceforth, increase in consensus time for every such message exchange further decreases the effective time utilization of SU throughput due to control packets.

From the channel end, free spectrum in the form of operating channels and CCC is analyzed in Fig. 13(a) and (b) respectively. As noted from Fig. 13(a), about 70–75% of free spectrum is available to SUs as the rest is occupied by control packets. Considering their effectiveness in reducing handoff delay and call drops, such tolerance can be allowed in CRN.

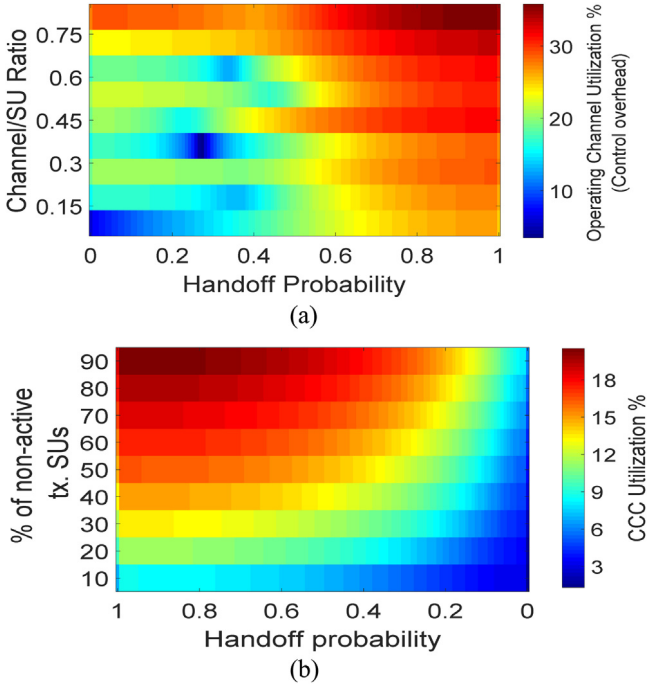


Fig. 13. Free spectrum utilization by control packets in (a) operating channel; and (b) CCC under different scenarios.

Finally, Fig. 13(b) reveals that only few cases correspond to CCC utilization. This is deliberately designed to keep CCC free for instant access in case SUs require them for reactive handoff. As evident from Fig. 13(b), a maximum of 19% utilization is recorded for CCC even when more than 90% of the total SUs are in handoff mode. This is due to lesser number of reactive handoff instances that can be further attributed to the generation of periodic TCS as well as incorporation of ML procedure.

In a nutshell, network control packets are essential to maintain VoIP QoS and network capacity. However, their overhead must be within tolerable limits and the same is confirmed from this section and further validated in the following section.

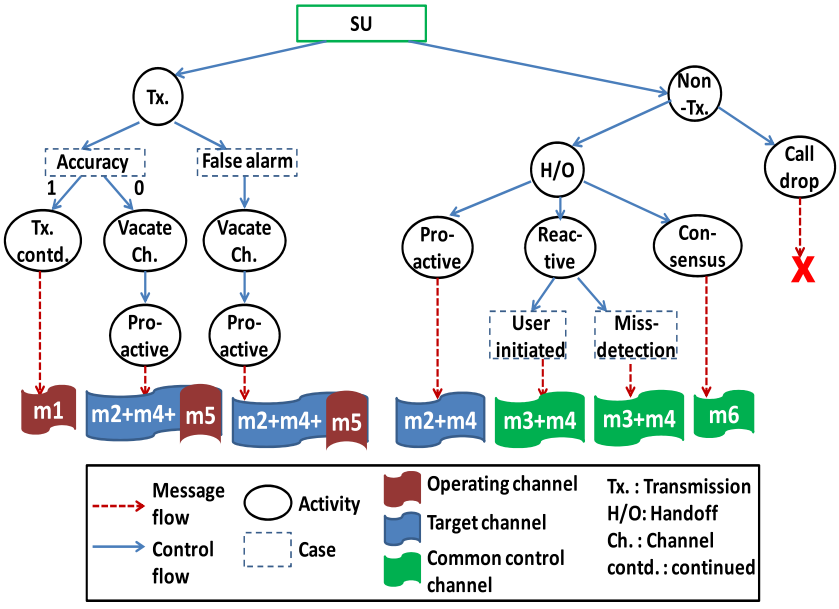


Fig. 12. Illustration of all cases involved in control packet overhead.

Table 7
TCS calculation by the SC terminal as per the proposed methodology in the test-bed.

Input: Channels	1	2	3	4	5	6	7	8	9	10
Probability Set	(4,3,3)	(5,4,1)	(7,2,1)	(8,1,1)	(5,3,2)	(3,3,4)	(2,3,5)	(4,4,2)	(6,3,1)	(1,2,7)
Measured PU Throughput (kbps)	2290	1769	1582	1520	1940	2609	3010	2054	1633	3642

Output: Calculated Channel Sorting List based on increasing throughput: {4,3,9,2,5,8,1,6,7,10}; Channel Sorted List based on the proposed method: {3,4,9,2,5,8,1,6,7,10}; TCS for first SU: {4,9,2,5}; TCS Values: Total Busy Time = 709 ms(SC), 740 ms(Actual); Total Idle Time = 3287 ms(SC), 2982 ms(Actual).

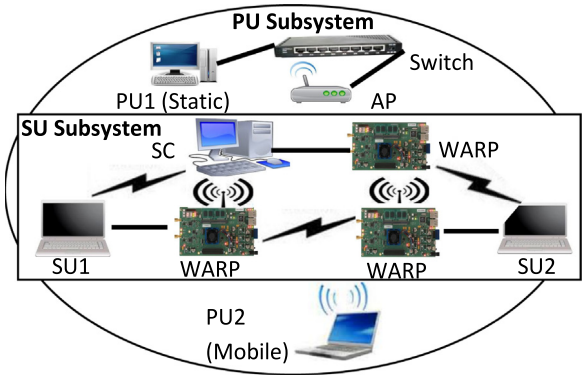


Fig. 14. Test-bed model with the PU subsystem and centralized SU subsystem.

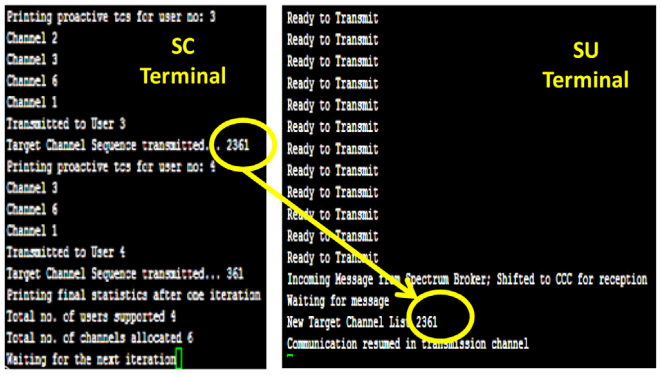


Fig. 15. Screenshots from WARP serial monitor illustrating successful channel allocation of periodic TCS by the SC terminal to the SU node.

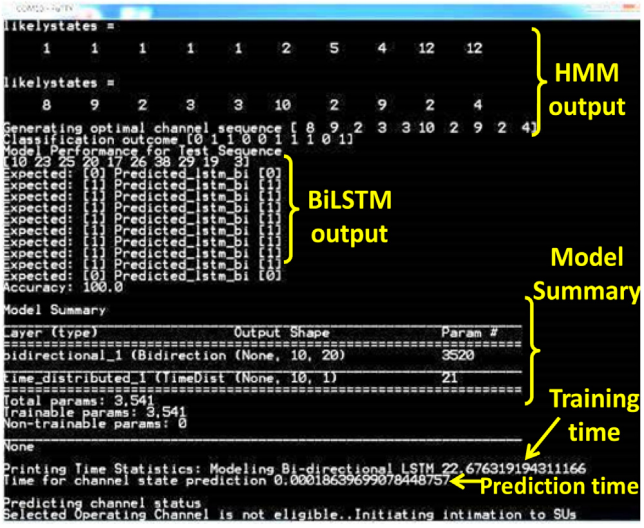


Fig. 16. Screenshots from WARP serial monitor illustrating implementation of the CSP block highlighting both HMM and BiLSTM methods.

7. Proof of concept: Test-bed implementation

This section validates the proposed work in terms of its practical applicability through test-bed design and execution.

7.1. Generic test-bed setup for the centralized CRN

A generic test-bed is developed as per the system model in Section 3 and is illustrated in Fig. 14. It comprises of SU and PU subsystems. SU subsystem includes SC and two SUs modeled using WARP kits (Wireless Open-Access Research Platform). CR functionalities are incorporated in these nodes by writing robust MAC and PHY codes in WARP SDK. VoIP communication is further enabled using H.323 protocol suite [78] and iLBC codec [78] having 13.33 kbps (33.33 pps) data rate.

Since WARP uses ISM (2.4 GHz) band, a separate Wi-Fi network is created for the PU subsystem using HP ProCurve switch and MSM310 Access Points. PU subsystem comprises of Wi-Fi enabled computers to perform data transmission. PU traffic follows exponential on–off model [82]. Channel conditions are varied as per Table 4. VQManager and Wireshark are used to monitor various statistics.

The SC node is configured with 3 ms sensing period. A low RSSI (Received Signal Strength Indicator) value of 2500 (−83.15 dBm threshold power) is used to detect the presence of any PU. A channel is sensed busy if the detected energy in WARP is greater than 0.0044pJ during this sensing period.

7.2. TCS calculation and allocation by the SC terminal

Table 7 demonstrates the calculation of proactive/periodic TCS. Ten channels are configured in the test-bed as per Table 4. The TCS is calculated based on the descending order of recorded PU throughput. It is noteworthy that the generated TCS under the proposed method closely resembles the calculated TCS in terms of channel sequence and total idle time. This validates the efficiency of the proposed TCS generation method.

A minor difference is noted between the calculated TCS and TCS generated by the proposed scheme as observed in Table 7. This is attributed to two reasons. Firstly, practical constraints in WARP limit its sensing accuracy. Secondly, SC generates TCS based on exponential on–off PU traffic as per the proposed method. However, actual on–off traffic values in the test-bed are varied with different probabilities as per Table 4.

Fig. 15 provides the screenshot during actual TCS allocation process in the test-bed. SC calculates and allocates unique TCS to every SU. On receiving periodic update message, SU temporarily shifts from the transmission channel to the CCC, receives periodic TCS and successfully resumes transmission. This confirms practical applicability of the proposed method.

7.3. Implementation of CSP block in SC terminal

Incorporation of CSP block in the SC terminal has been a critical challenge particularly because WARP is not capable of handling complex integration of HMM and BiLSTM methods. It is therefore realized using two components – (i) a server computer with rich hardware specifications (64 GB RAM, NVIDIA GeForce GTX 970 graphics, Intel

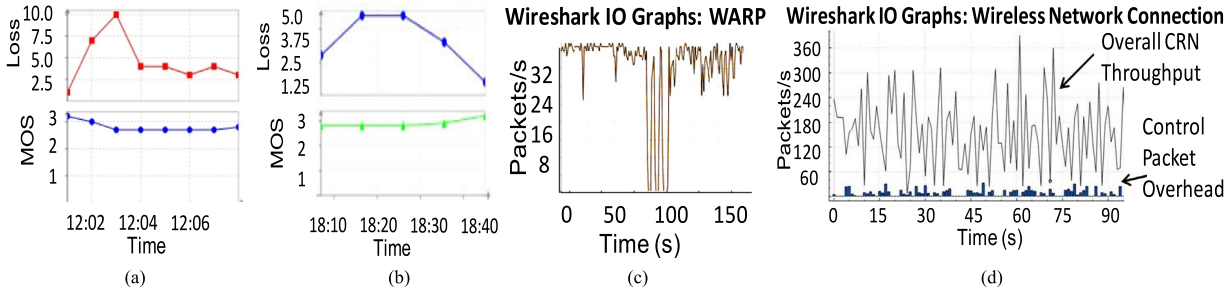


Fig. 17. (a) QoS readings depicting loss% and MOS from VQManager under (a) PROLOC; and (b) PROLOC-ML framework; Wireshark recordings depicting (c) SU throughput and per-instance spectrum handoff delay; and (d) control packet overhead over the total SU throughput.

Core i5-6600K overclocked processor, ASUS Z170 motherboard); and (ii) PuTTY (a free SSH and Telnet Client).

The server is connected to WARP via PuTTY at one end. In addition, it is configured with VoIP proxy server features enabling it to record MOS and generate HMM matrices. MATLAB is used to implement the Viterbi based HMM algorithm on this data and is further imported to Python environment which executes the BiLSTM algorithm. Finally, PuTTY is fed with the channel eligibility output by the server and is called by WARP kit to connect to the SC terminal.

Fig. 16 provides a screenshot of the aforementioned operation. It is observed that as per the underlying hardware, the model takes 22.67 s to build based on HMM output. However, channel prediction works real-time incurring only 186.39 μs and making it apt for the current application. This time can be further reduced by increasing the hardware specifications for SC. Moreover accuracy of more than 99% is obtained after training the model for 250 epochs based on the output from Fig. 9. Model functionality including HMM output, training and testing phases and overall model summary as per Fig. 3 is also provided as output (although it is not required practically). Finally channel eligibility decision is provided by PuTTY so that SC can timely intimate the SUs.

7.4. Spectrum handoff operation by the VoIP SU

VoIP QoS metrics are recorded with respect to PROLOC and PROLOC_ML methodologies in Fig. 17(a) and (b) respectively. VQManager readings in Fig. 17(a) record a minimum MOS value of 2.7, maximum 10% packet loss for SU and negligible PU packet loss ratio (= 0.005). PROLOC_ML further reduces packet loss to less than 5% which is attributed to lesser number of handoff instances. Thus, this work reduces QoS degradation for SUs during handoff while protecting PU traffic from interference.

Spectrum handoff delay is affected by device impairment delays in WARP. However, it is still below 1.8 s as recorded in Fig. 17(c). As this delay is below the RTCP call time-out value, there is no call drop. Moreover, call resumption in the target channel implies successful channel allocation by the SC and subsequent consensus achieved by two VoIP SUs.

Finally network control overhead is measured using protocol filtering mechanism in Wireshark. While VoIP packets are transmitted over RTP/UDP, TCP is used for control packet transmission. Thus different protocols facilitate in packet filtering. It is evident from Fig. 17(d) that the mean ratio of network control overhead to total throughput is 20–25%. This verifies the inference drawn from simulation in Fig. 13(a). Furthermore, improved VoIP QoS under PROLOC-ML suggests that this control overhead does not affect VoIP QoS.

Thus, test-bed studies establish the credibility of the proposed scheme in sustaining real-time communication during spectrum handoff. A complete video demonstration of the test-bed operation is also provided as proof of concept.

Finally in the last section, the overall performance gain achieved by the spectrum handoff enabled VoIP based CRN over conventional networks is discussed.

7.5. Comparative analysis with conventional network

A basic network as referred to in this section is a conventional one that does not follow the overlay principle of CRN. It implies that instead of opportunistic communication, this network allows different users to utilize idle channels using multiple access procedure. This section compares such a basic network with the one developed in this paper. The aim is to establish the business prospect of our designed system in emerging systems.

To this end, sum goodput as introduced in [83] is selected as the primary performance metric. Sum goodput is the average summation of the entire throughput generated by different combinations of active users in the network. For a basic network having capacity of N users and i channels, let j users be active with probability p and transmit with data rate R_b. The corresponding sum goodput S_b is expressed as follows.

$$S_b = \sum_{j=1}^{N} \binom{N}{i} p^j (1-p)^{N-j} C_b \quad where \quad C_b = R_b j (1-1/i)^{j-1} \tag{36}$$

VoIP based CRN as explored in this paper offers two added benefits over a basic network – (1) licensed users provide access to unlicensed VoIP SUs for transmission during their idle periods as per overlay CRN policy; (2) under on–off traffic accompanied by silence suppression features of VoIP codecs, data SUs may occasionally utilize these channels further as explored in [84]. Accordingly sum goodput S_{cr} for VoIP based CRN is derived in (37) where the notations are described as follows.

R_v, R_d = data rate of VoIP and data SUs respectively; j', j'' = number of VoIP and data SUs respectively; i', i'' = total number of channels available to VoIP and data SUs respectively; $\Pr_{drop_v}$, $\Pr_{drop_d}$ = dropping probability for VoIP and data SUs respectively; j_{cr} = total number of active users.

Obviously, system capacity is increased for S_{cr} as j_{cr} is greater than j. Additionally, once our proposed TCS model is implemented, the capacity even increases as SUs instead of dropping permanently find other suitable channels to resume their transmission. Accordingly, C_v in (37) is modified to $C_{v_h|o}$ that incorporates the spectrum handoff enabled SUs in CRN. $C_{v_h|o}$ is given by (38). See Eq. (37) given in Box II

$$\begin{aligned} C_{v_h|o} = {} & R_v j' \left[\left(1 - \Pr_{busy}^{(w_{i'})}\right) f(i') + \Pr_{busy}^{(w_{i'})} \left[\left(1 - \Pr_{busy}^{(w_{i'-1})}\right) f(i'-1) \right. \right. \\ & + \Pr_{busy}^{(w_{i'-1})} \left[\left(1 - \Pr_{busy}^{(w_{i'-2})}\right) f(i'-2) + \Pr_{busy}^{(w_{i'-2})} \left[\left(1 - \Pr_{busy}^{(w_{i'-3})}\right) f(i'-3) \right. \right. \\ & \left. \left. \left. + \cdots (n-1)\ times \cdots + \Pr_{busy}^{(w_{i'-(n-1)})} \left[\left(1 - \Pr_{busy}^{(w_{i'-n})}\right) f(i'-n) \right] \right] \cdots \right] \\ & \text{where } f(i') = (1 - 1/i')^{j'-1} \end{aligned} \tag{38}$$

All three networks namely, the basic network, VoIP based CRN and spectrum handoff enabled CRN are implemented in simulation (Fig. 18(a)) and test-bed (Fig. 18(b)). It is seen that both simulation and hardware platforms record almost double the throughput for VoIP based CRN as compared to a basic network. The increase is manifold

$$S_{cr} = \sum_{j_{cr}=1}^{N} \binom{N}{i} p^{j'} (1-p)^{N-j'} \left[C_b + H_{cr} \left(1 - \mathrm{Pr}_{drop_v}\right) \left(C_v + H_{ss} \left(1 - \mathrm{Pr}_{drop_d}\right) C_d \right) \right]$$

where $C_v = R_v j' \left(1 - 1/i'\right)^{j'-1}$; $C_d = R_d j'' \left(1 - 1/i''\right)^{j''-1}$; $j_{cr} = j + j' + j''$; $i'' \subseteq i' \subseteq i$;

$H_{cr} = 0$ if SUs are allowed to operate as part of overlay CRN
$= 1$ no SU is allowed
$H_{ss} = 0$ if silence suppression enabled codec is allowed
$= 1$ no data user further allowed during VoIP transmission

(37)

Box II.

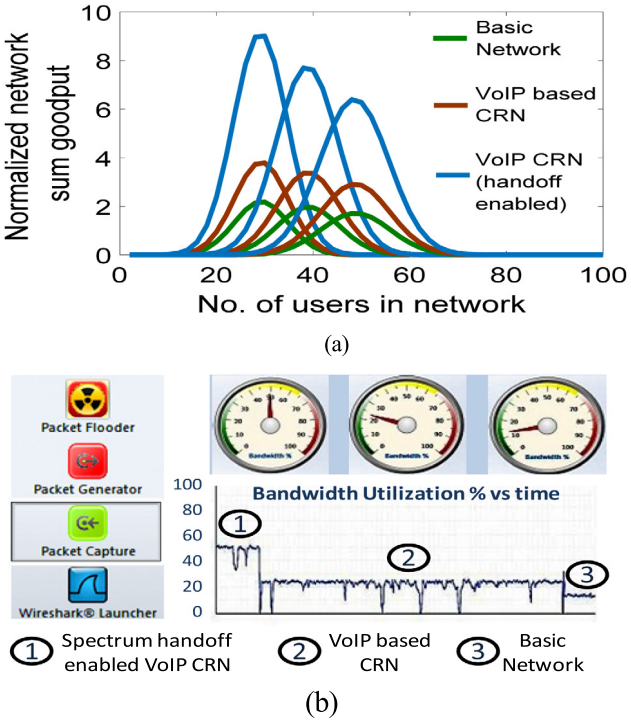


Fig. 18. Performance gain in terms of throughput of spectrum handoff enabled VoIP based CRN over other networks in (a) simulation; and (b) test-bed.

when the proposed TCS based spectrum handoff is enabled in CRN. This drastic increase is due to the fact that unlike a conventional VoIP based CRN, TCS based CRN further reduces the number of call drops thus increasing system capacity. An interesting observation from Fig. 2 is that an increase in number of users reduces sum goodput in all cases. This is because with more users, channel contention increases. Hence, the probability of an idle channel for SU access decreases and this effect is also realized during handoff instances.

Thus from the systems perspective, the proposed three-phase TCS scheme based CRN with VoIP SUs provide excellent motivation to the stakeholders for application deployment.

8. Conclusion

In this paper, we have proposed a ML-enabled three-phase TCS generation and allocation scheme to address the challenges in a multi-user CRN. The major contribution of this work is the formulation of novel solutions that provide time-bound generation and fair allocation of unique TCS to multiple real-time SUs along with intelligent prediction of channel states. With special focus on QoS aware VoIP applications for SUs, the proactive phase applies greedy technique to solve the fractional knapsack based TCS calculation problem in O (n log n) time. Optimal target channel selection is achieved by SC in terms of maximum cumulative channel idle time and minimum number of algorithmic iterations. The reactive phase prevents call drops by providing SUs with pseudo-random sequence of target channels when proactive TCS is rendered insignificant under dynamic channel conditions. The periodic phase timely updates the TCS as well as dynamically maintains CCC with the help of user, channel and TCS allocation vectors. The CSP block of the SC further reduces the worst-case reactive TCS scenarios through SC directed spectrum handoff operation by SUs. It includes a novel joint HMM-BiLSTM framework that learns VoIP performance with respect to channel states and uses ML technique to predict operating channel conditions. Studies in the formulated analytical framework confirm a drastic reduction in channel access conflict among multiple SUs, thus ensuring fair allocation of unique TCS. Comparative analysis in simulation further witnesses over 40% drop in average handoff delay as well as 45% decline in call drop probability under the proposed scheme, thereby recording the highest transmission duration per target channel. An accuracy of over 97% is obtained in channel prediction thereby marking its suitability under diverse channel conditions. Finally, successful test-bed execution validates the proposed methodology and establishes both its practical applicability of supporting real-time users in a multi-user CRN as well as the business prospect of deploying such networks for emerging systems and applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Tamal Chakraborty: Conceptualization, Methodology, Writing - original draft. **Iti Saha Misra:** Supervision, Validation, Writing - review & editing.

Appendix A. Determination of frame duration value – T_v

VoIP traffic for a particular SU is modeled as a simple on–off model. For a CRN having total N_v number of SUs, the average number of active SUs is given by (39). Each SU is active with probability Pr_{on} having mean on and off periods distributed exponentially and denoted by $1/\alpha_v$ and $1/\beta_v$ respectively. See Eq. (40) given in Box III.

$$j' = N_v \times \mathrm{Pr}_{on} \quad where \quad \mathrm{Pr}_{on} = \alpha_v^{-1} / \left(\alpha_v^{-1} + \beta_v^{-1}\right) \tag{39}$$

From systems perspective, VoIP traffic in CRN is modeled as a two-state MMPP (Markov Modulated Poisson Process) model [85] that captures the interdependency between VoIP frames. Let s_1 and s_2 denote the two states of MMPP for which the mean arrival rates following Poisson distribution are provided by λ_{s1} and λ_{s2} respectively. Additionally, the steady state probability of 1-D Markov chain modeling N_v VoIP SUs is given by,

$$\pi_k = \binom{N_v}{k} \mathrm{Pr}_{on}{}^{k} \left(1 - \mathrm{Pr}_{on}\right)^{N_v - k} \tag{41}$$

$$T_{su} = \sum_{m} \left[\left(1 - \mathrm{Pr}_{drop}^{m-1} \right) \times \left[\left[\left(1 - \mathrm{Pr}_{busy(ls)}^{(i)} \right) \times t_{hd}^{(j)} \right] + \left[\mathrm{Pr}_{busy(ls)}^{(i)} \times \left(1 - \mathrm{Pr}_{busy(hs)}^{(i+1)} \right) \times t_{ld}^{(j)} \right] \right] \right] \tag{40}$$

Box III.

Using IDC (Index of Dispersion for Counts) [86] to match MMPP model coefficients with that of the VoIP on–off model, the frame duration is derived as follows.

$$T_v = \left(\frac{\sum_{i=0}^{j'} (i \times \pi_i)}{\sum_{j=0}^{j'} \pi_j} + \frac{\sum_{i=j'+1}^{N_v} (i \times \pi_i)}{\sum_{j=j'+1}^{N_v} \pi_j} \right) \Big/ \left(\lambda_{s1} + \lambda_{s2} \right) \tag{42}$$

Based on this frame duration value, SC can easily estimate the average transmission slot required for SU by simply multiplying T_v with a predefined number of frames.

Appendix B. Determination of SU transmission duration – T_{su}

Since this paper deals with SC initiated TCS operations, SU related operations including sensing and transmission have not been discussed as they are out of scope for this work. Briefly stating, SUs in our test-bed follow an adaptive sensing and transmission strategy. It consists of two sensing and transmission durations namely t_{ls} and t_{ld} for short sensing and transmission period respectively and t_{hs} and t_{hd} for the longer counterpart. SU initially starts with t_{ls} slot. If channel is idle, it transmits for t_{hd} slot before next sensing occurs. If channel is busy, SU senses for another t_{hs} slot. If the sensing outcome is finally idle, SU transmits for t_{ld} period. For two successive busy outcomes with t_{hs} slot, SU discards that channel.

Based on this policy, the time duration T_{su} for a period of m cycles is derived in Eq. (40) where the different components are expressed as follows.

$$\mathrm{Pr}_{drop}^{m} = \mathrm{Pr}\left[\sum_{k=1}^{n_{ls}} H_k \geq H_{th} \right]_{i-2} \times \prod_{l=i-1,i} \mathrm{Pr}\left[\sum_{k=1}^{n_{hs}} H_k \geq H_{th} \right]_{l} \tag{43}$$

$$n_{ls} = \frac{t_{ls}}{p}; \tag{44}$$

$$n_{hs} = \frac{t_{hs}}{p} \tag{45}$$

$$\mathrm{Pr}_{busy(ls)} = \mathrm{Pr}\left[\sum_{k=1}^{n_{ls}} H_k \geq H_{th} \right] \tag{46}$$

$$\mathrm{Pr}_{busy(hs)} = \mathrm{Pr}\left[\sum_{k=1}^{n_{hs}} H_k \geq H_{th} \right] \tag{47}$$

Obviously, SC is not aware of such adaptive SU policies. Therefore, estimation of transmission duration by SC varies from the actual transmission slot for SU.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.comcom.2020.02.026.

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